

———— // ZPWR-FX // ————

zpwr-fx

v0.1.0

MODULE REFERENCE

wire your own algorithm from primitive DSP blocks

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MenkeTechnologies

———— MENKETECHNOLOGIES ————

Signal Sources

External sources you can cable into any block input or modulation route.

1 In L 2 In R 3 Noise Mod Wheel Aftertouch Pitch Bend Velocity Expression Breath Sustain Note MPE Press MPE Slide MPE Bend

Oscillator

33 modules

Wavetable 0 inputs

Wavetable oscillator: it advances a phase at Freq and reads single-cycle frames loaded from the node config, interpolating both within and between frames. Position (0 to 1) morphs across the frame stack, Warp (-1 to +1) phase-distorts the read pointer for timbral bending, Freq sets pitch (20 to 8000 Hz), and Level scales output. A pure generator (no inputs) that sweeps from one waveform to another as Position moves.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Position	0 – 1	0	—
Warp	-1 – 1	0	—
Level	0 – 1	0.5	—

Granular 1 input

Granular oscillator/texture: continuously captures the input into a ~2 s ring buffer and sprays overlapping Hann-windowed grains read back from it. Size sets grain length in ms, Density the spawn rate in grains/sec, Pitch transposes each grain in semitones, and Jitter randomizes the read origin for a diffuse, smeared texture. Mix blends the grain cloud against the dry input -- low density gives a sparse stutter, high density a thick wash.

Param	Range	Default	Unit
Size	5 – 500	120	ms
Density	1 – 200	40	—
Pitch	-24 – 24	0	st
Jitter	0 – 1	0.3	—
Mix	0 – 1	1	—

Spectral 1 input

FFT spectral-resynthesis oscillator: a streaming 1024-pt STFT phase vocoder (75% overlap) that FFTs each frame, reshapes the spectrum, then iFFT's. Stretch scales the spectral envelope by resampling bins, Shift moves bins up/down for a frequency shift, and Blur smears each bin's magnitude over time. Mix crossfades the resynthesis against the dry signal -- pushes toward metallic, harmonically reshaped tones.

Param	Range	Default	Unit
Blur	0 – 1	0.3	—
Shift	-50 – 50	0	bin
Stretch	0.5 – 2	1	—
Mix	0 – 1	1	—

Susaw 0 inputs

Massive X super-saw: sums 7 saws spread across a detune range, with a slow random drift added to the base frequency, then scaled by Level. Detune sets the +/-3% spread between voices, Blend lowers the 6 side voices relative to the loud centre voice, and Drift

sets how much the wandering frequency offset is applied. Higher Blend gives a wider unison wash; lower Blend keeps the centre pitch dominant for a tighter lead.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Detune	0 – 1	0.4	—
Blend	0 – 1	0.7	—
Drift	0 – 1	0.1	—
Level	0 – 1	0.5	—

Sampler inputs

Reads one of the 8 shared SampleStore slots with linear interpolation, advancing a playback position scaled by pitch and the slot/host sample-rate ratio, then scaled by Level. Slot picks the source, Pitch transposes -24 to +24 semitones, Start sets the entry point (0 to 0.99 of the buffer), and Loop chooses whether playback wraps back to Start or stops at the end. Useful for looped wavetable-style sources or one-shot sample playback depending on Loop.

Param	Range	Default	Unit
Slot	0 – 7	0	—
Pitch	-24 – 24	0	st
Start	0 – 1	0	—
Loop	0 – 1	1	—
Level	0 – 1	0.5	—

Sync inputs

Hard-sync oscillator: a slave saw runs at Sync times the master frequency and is reset to zero each time the master phase wraps, producing the classic sync-sweep timbre, then scaled by Level. Freq sets the master pitch and Sync sets the slave ratio (1 to 6). Higher Sync ratios add brighter, more formant-like harmonics from the forced phase resets.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Sync	1 – 6	2	—
Level	0 – 1	0.5	—

Warp inputs

Phase-warping oscillator: a master phase feeds a base waveform but is reshaped by one of several Mode warps (Neutral, Wrap, Grain, Hardsync, Hard Up-Down, Bend, Mirror, Random, Jitter) before the wave is read. Wave picks the base shape (sine, saw, square, triangle), Mode selects the warp, Amount sets warp depth, and Ratio scales the slave frequency for the sync modes. Pushing Amount drives anything from subtle phase bend to per-sample jitter noise depending on Mode.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Wave	0 – 3	1	—
Mode	0 – 8	0	—
Amount	0 – 1	0.5	—
Ratio	1 – 8	2	—

Sub inputs

Dedicated sub oscillator: runs a single phase at Freq dropped by Octave, outputting either a sine or a square, then scaled by Level. Octave shifts down 0, 1, or 2 octaves, Wave selects sine or square, and Level sets the output gain. Use it to add low-end weight under a main oscillator, with sine for clean sub and square for a harder, more present low end.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Octave	-2 – 0	-1	—
Wave	0 – 1	0	—
Level	0 – 1	0.6	—

Spire inputs

Spire-style multimode oscillator: sums up to 7 detuned unison voices, each rendered by the selected Mode (Classic saw, Noise, FM, AMSync, SawPWM, HardFM, Vowel) with the sum normalized by the voice count. Mode picks the algorithm, Shape is the per-mode control (FM index, sync ratio, pulse width, or vowel position), Voices sets unison count, and Detune spreads the voices over a +/-3% range. The Vowel mode adds three morphing formant band-passes over the summed saw for talking-synth timbres.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Mode	0 – 6	0	—
Shape	0 – 1	0.5	—
Voices	1 – 7	1	—
Detune	0 – 1	0.2	—

PMOsc inputs

Phase-modulation oscillator: a sine modulator running at Ratio times Freq offsets the carrier phase by depth Index, and the warped phase is read through the selected carrier waveform, then scaled by Level. Wave selects the carrier shape (Sine, Tri, three bent/saturated triangle variants, narrowed sine), Ratio sets the modulator-to-carrier frequency, and Index sets modulation depth. Raising Index and Ratio builds increasingly bright, inharmonic FM-style spectra.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Wave	0 – 5	0	—
Ratio	0.5 – 8	1	—
Index	0 – 8	2	—
Level	0 – 1	0.5	—

PhaseDist inputs

Casio CZ phase-distortion oscillator: a phase ramp drives a cosine through a warped read so DCW morphs a sine toward the chosen shape, then scales by Level. Shape selects Saw or Pulse (true phase distortion) or three resonant modes (ResoSaw, ResoTri, ResoTrap) that are windowed hard-sync, and DCW sets the distortion amount or, in resonant modes, the resonant multiple (1 to 16). Low DCW stays near a clean sine while high DCW opens into bright sawtooth, pulse, or formant-like resonant tones.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Shape	0 – 4	0	—
DCW	0 – 1	0.5	—
Level	0 – 1	0.5	—

Geiger inputs

Sparse-impulse noise: each sample has a probability set by Rate of firing a random-amplitude impulse, which then decays exponentially, all scaled by Level. Rate sets the average clicks per second and Decay sets how long each impulse rings before fading. Low Rate with short Decay gives sparse ticks like a Geiger counter, while higher Rate and longer Decay thicken it toward a crackling texture.

Param	Range	Default	Unit
Rate	0.1 – 200	20	Hz
Decay	0 – 1	0.5	—
Level	0 – 1	0.5	—

Multisampler 3 inputs

Multisampler: on each note-on (from the note, gate, and velocity inputs) it strikes every region whose key and velocity zones match, resampling each by note-vs-root plus fine tune so it plays at the right pitch. Slot selects which loaded .sfz region set to use (0 to 7), Loop forces looping of the regions' loop or play windows, and Level scales output; velocity crossfades and round-robin sequencing come from the region data. Use it for realistic multisampled instruments where timbre tracks key and velocity.

Param	Range	Default	Unit
Slot	0 – 7	0	—
Loop	0 – 1	0	—
Level	0 – 1	0.6	—

Pluck 2 inputs

Karplus-Strong plucked-string physical model: a gate edge fills a delay line with a white-noise burst, then each pass averages neighbouring samples and scales by Decay so the noise settles into a pitched, decaying tone. Tune shifts the note in semitones (sets the delay length from the in0 note), Damp blends in more of the lowpass average to dull the highs, Decay sets how slowly the string rings out, and Level scales the output. Plucks on the in1 gate; short Decay and high Damp give a soft muted pluck, long Decay and low Damp a bright sustained string.

Param	Range	Default	Unit
Tune	-24 – 24	0	st
Damp	0 – 1	0.5	—
Decay	0.8 – 0.999	0.99	—
Level	0 – 1	0.6	—

Modal 1 input

Modal synthesis: a bank of 6 tuned decaying resonators struck by a gate (bells, mallets, struck metal). Structure morphs harmonic to inharmonic-bar partials.

Param	Range	Default	Unit
Freq	20 – 4000	220	Hz
Decay	0.05 – 8	1.5	s
Structure	0 – 1	0.3	—
Bright	0 – 1	0.6	—
Level	0 – 1	0.6	—

Blown 1 input

Waveguide wind instrument (STK-clarinete): a bore delay line with a nonlinear reed reflection, driven by breath from the gate. The blown counterpart to Karplus-Strong.

Param	Range	Default	Unit
Freq	20 – 2000	220	Hz
Breath	0 – 1	0.5	—
Reed	0 – 1	0.5	—

Param	Range	Default	Unit
Tone	0 – 0.9	0.4	—
Level	0 – 1	0.6	—

Bowed 1 input

Bowed-string waveguide (STK-bowed): two delay lines for the string halves with a stick-slip bow-table at the bow point, driven by the gate. Violin/cello family.

Param	Range	Default	Unit
Freq	20 – 2000	220	Hz
Force	0 – 1	0.5	—
Pos	0.02 – 0.5	0.12	—
Tone	0 – 0.95	0.5	—
Level	0 – 1	0.6	—

FOF 0 inputs

Formant-wave-function synthesis (CHANT): each fundamental period retriggers a bank of decaying formant grains, generating vocal/formant tones. Vowel morphs A-E-I-O-U.

Param	Range	Default	Unit
Freq	20 – 800	110	Hz
Vowel	0 – 4	0	—
Bandwidth	0.2 – 4	1	—
Level	0 – 1	0.5	—

WaveTerrain 0 inputs

Wave-terrain synthesis: a Lissajous orbit traverses a 2D polynomial surface $z=f(x,y)$; the height is the output. Warp tilts the orbit, Ratio sets its y/x frequency.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Ratio	0.5 – 8	1	—
Warp	0 – 1	0	—
Level	0 – 1	0.5	—

Pulsar 0 inputs

Pulsar synthesis (Roads): a pulsaret (windowed sine) is emitted once per fundamental period, then silence. Freq sets pitch, Formant the pulsaret cycles, Width the duty.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Formant	1 – 16	3	—
Width	0.05 – 1	0.5	—
Level	0 – 1	0.5	—

Scanned 1 input

Scanned synthesis (Mathews/Verplank): a 16-mass spring ring evolves slowly after a gate hammer-strike; a scan reads the mass positions at audio rate as the waveform. Tension/Damping shape the timbre's evolution.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Tension	0.05 – 1	0.4	—
Damping	0 – 1	0.2	—
Level	0 – 1	0.6	—

Noise 0 inputs

Colored-noise generator: Color morphs from white (flat spectrum) toward a low-passed pink-ish tone, scaled by Level. Use it as a percussion/wind source, a sample-and-hold input, or to add air and dither. It takes no signal input - it is a pure generator.

Param	Range	Default	Unit
Color	0 – 1	0	—
Level	0 – 1	1	—

Oscillator 0 inputs

Audio-rate tone generator with sine, saw, square and triangle waves (Wave) at Freq. Use it as a ring-mod or FM carrier, a drone, a sub layer, or a test tone. It takes no signal input - patch its output into RingMod, a filter, or straight to Out.

Param	Range	Default	Unit
Freq	20 – 8000	220	Hz
Wave	0 – 3	0	—
Level	0 – 1	0.5	—

VAnalog 0 inputs

Virtual-analog oscillator: saw or pulse (Wave) with adjustable pulse width (PWM), a square sub-oscillator an octave down (Sub), and slow random Drift for analog pitch instability. Level scales the output. A warm, slightly unstable VA voice - pair with the Mini/Jup-8 analog filters for a classic synth chain.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz
Wave	0 – 1	0	—
PWM	0.05 – 0.95	0.5	—
Sub	0 – 1	0	—
Drift	0 – 1	0.1	—
Level	0 – 1	0.5	—

FM 0 inputs

Two-operator FM voice (the DX7 building block): a sine carrier is phase-modulated by a sine modulator at Ratio times the frequency, with Index setting depth and modulator self-Feedback adding harmonic edge toward noise. Integer ratios give harmonic, organ/bass timbres; non-integer ratios give bells and metallic tones.

Param	Range	Default	Unit
Freq	20 – 8000	110	Hz

Param	Range	Default	Unit
Ratio	0.5 – 8	2	—
Index	0 – 12	2	—
Feedback	0 – 0.9	0	—
Level	0 – 1	0.5	—

Additive inputs

Additive oscillator that sums up to 16 sine partials (Partials) with a spectral Tilt: negative Tilt rolls off the upper harmonics for a soft, mellow tone, positive brightens toward a buzzy saw. Partials above Nyquist are skipped, so it stays alias-free. A clean, controllable harmonic source from pure sine to bright buzz.

Param	Range	Default	Unit
Freq	20 – 4000	110	Hz
Partials	1 – 16	8	—
Tilt	-2 – 2	-0.7	—
Level	0 – 1	0.5	—

Organ inputs

Additive drawbar organ oscillator that sums four sine partials at the first four harmonics of Freq. Bar1, Bar2 and Bar3 set the drawbar levels of the fundamental, second and third harmonics (the fourth harmonic tracks Bar2 at half level), and Level sets the overall output. A self-running tone generator with no audio input, voiced from sine-only partials for a clean drawbar timbre.

Param	Range	Default	Unit
Freq	20 – 4000	220	Hz
Bar1	0 – 1	1	—
Bar2	0 – 1	0.6	—
Bar3	0 – 1	0.4	—
Level	0 – 1	0.5	—

ColorNoise inputs

Noise generator producing white, pink or brown noise from an xorshift source filtered to taste. Color selects the spectrum: flat white, a one-pole pink-tilted blend, or a heavily integrated brown noise with a steeper low-frequency emphasis. Level sets the output amplitude, making it a self-running source for textures, test signals or modulation.

Param	Range	Default	Unit
Color	0 – 2	1	—
Level	0 – 1	0.5	—

Pulsaret inputs

Pulsar synthesis oscillator: at each fundamental period a Hann-windowed sinusoidal grain (the pulsaret) is retriggered, with silence filling the rest of the cycle. Freq sets the fundamental (grain repetition rate), Formant sets the sine frequency inside the grain (formant pitch), Width sets the fraction of the period the grain occupies, and Level scales the output. Narrow widths and high formants give buzzy, formant-rich tones.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Formant	1 – 16	4	—
Width	0.1 – 1	0.5	—

Param	Range	Default	Unit
Level	0 – 1	0.5	—

WaveScan 0 inputs

Additive harmonic scanner: sums a series of sine partials whose amplitudes follow a power-law roll-off, then normalizes the result. Freq sets the fundamental, Harmonics sets how many partials are stacked, Scan tilts the spectrum from bright (all harmonics equal at 0) toward dark (fundamental only as it approaches 1), and Level scales the output. Sweep Scan for a sawtooth-to-sine timbral morph.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Scan	0 – 1	0.3	—
Harmonics	1 – 16	8	—
Level	0 – 1	0.5	—

SuperPWM 0 inputs

Supersaw built from detuned pulse waves: stacks several pulse oscillators spread symmetrically in tuning and averages them for a thick unison. Freq sets the center pitch, Detune spreads the voices apart in tuning, Width sets the pulse duty cycle (timbre), Voices sets how many oscillators stack, and Level scales the output. High detune and voice counts give wide, rich pad and lead tones.

Param	Range	Default	Unit
Freq	20 – 2000	110	Hz
Detune	0 – 1	0.3	—
Width	0.05 – 0.95	0.5	—
Voices	1 – 7	5	—
Level	0 – 1	0.5	—

BellFM 0 inputs

Two-operator FM bell: a sine carrier is phase-modulated by a sine modulator running at a non-integer multiple of the carrier frequency, producing inharmonic metallic partials. Freq sets the carrier pitch, Ratio sets the modulator-to-carrier frequency ratio (typically fractional for bell tones), Index sets the modulation depth (brightness and partial spread), and Level scales the output. Higher Index and non-integer Ratio yield clangorous, bell-like timbres.

Param	Range	Default	Unit
Freq	20 – 2000	220	Hz
Ratio	0.5 – 8	3.5	—
Index	0 – 8	3	—
Level	0 – 1	0.5	—

Wind 0 inputs

Wind texture from filtered noise: white noise is fed through a resonant TPT band-pass whose center frequency is modulated by a slow 0.3 Hz gust LFO. Cutoff sets the band-pass center frequency, Reso sets the resonance (how whistling/pitched the noise becomes), Gust scales the LFO cutoff sweep depth, and Level scales the output. Produces ambient wind, breath and atmospheric beds.

Param	Range	Default	Unit
Cutoff	100 – 8000	800	Hz
Reso	1 – 20	8	—

Param	Range	Default	Unit
<code>Gust</code>	0 – 1	0.5	—
<code>Level</code>	0 – 1	0.5	—

Filter

51 modules

Vocoder 2 inputs

Analysis/synthesis vocoder: splits the carrier at in0 and modulator at in1 into log-spaced bandpass bands, tracks each modulator band envelope, and multiplies the matching carrier band by it before summing, all blended by Mix against the dry carrier. Bands sets the number of analysis bands (4 to 16), Q sets band sharpness, and Attack sets the envelope follower attack time in ms. More bands and higher Q give more intelligible speech imprinting, while shorter Attack tracks transients more tightly.

Param	Range	Default	Unit
Bands	4 – 16	12	—
Q	1 – 16	6	—
Attack	1 – 50	5	ms
Mix	0 – 1	1	—

PZFilter 1 input

Drawable magnitude-response filter: samples a frequency-to-gain curve drawn in node config at 16 log-spaced bands and bakes one peaking-EQ biquad per band, passing the input at in0 through the series to reproduce the drawn shape. Amount morphs between the dry input and the fully filtered output. Draw boosts and cuts across the spectrum to build custom EQ curves or resonant filter shapes.

Param	Range	Default	Unit
Amount	0 – 1	1	—

PeakFilter 1 input

A biquad peaking EQ band that boosts a single frequency region, followed by a tanh soft limiter to tame the boosted peaks. Freq sets the center frequency (40..16000 Hz), Gain sets the boost amount (0..18 dB), Edge sets the band Q/sharpness, and Mix blends the filtered signal against dry. Useful for emphasizing a resonant frequency without hard clipping when pushed.

Param	Range	Default	Unit
Freq	40 – 16000	1000	Hz
Gain	0 – 18	9	dB
Edge	0.5 – 12	3	Q
Mix	0 – 1	1	—

FilterLFO 1 input

A 4-pole resonant ladder lowpass swept by an internal LFO that morphs between triangle and sawtooth shapes. Rate sets the LFO speed (Hz), Range sets the sweep depth around the cutoff (~60 Hz..15.6 kHz), Reso sets the resonant feedback, Shape morphs the LFO from triangle to saw, and Mix blends wet against dry. High Reso adds squelch while Shape changes the sweep from symmetric to ramped.

Param	Range	Default	Unit
Rate	0.02 – 12	0.5	Hz
Range	0 – 1	0.7	—
Reso	0 – 1	0.5	—
Shape	0 – 1	0	—
Mix	0 – 1	1	—

LowPassGate 2 inputs

Buchla west-coast low-pass gate: a vactrol couples a VCA and VCF so a ping opens both with the characteristic nonlinear ringing decay. Mode picks VCA / VCF / both.

Param	Range	Default	Unit
Ping	0.01 – 2	0.4	s
Color	0 – 1	0.5	—
Mode	0 – 2	2	—
Level	0 – 1	0.8	—

SplitEQ 1 input

A peaking EQ that knows transients from tone: a fast/slow envelope split weighs the band gain toward the attack or the sustain, so you can brighten transients without harshening the body.

Param	Range	Default	Unit
Freq	40 – 16000	2000	Hz
Transient	-18 – 18	4	dB
Tonal	-18 – 18	0	dB
Q	0.3 – 8	1.5	—
Mix	0 – 1	1	—

Mini 3 inputs

Minimoog-style 4-pole transistor ladder: per-stage saturation thins the bass as resonance climbs to self-oscillation.

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Jup-8 3 inputs

Jupiter-8-style 4-pole OTA ladder: smooth resonance that keeps the low end (no Moog bass loss).

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

MS-20 3 inputs

MS-20-style 2-pole Sallen-Key low-pass: diode clipping in the resonance path screams and distorts at high resonance.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

SEM 3 inputs

Oberheim SEM-style 2-pole state-variable filter: continuously morphs low-pass through notch to high-pass.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Morph	0 – 1	0	—

EMS 3 inputs

EMS VCS3-style 3-pole (18 dB) diode ladder: asymmetric diode shaping gives a sharp, acidic resonance.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Wasp 3 inputs

EDP Wasp-style 12 dB multimode filter: CMOS-inverter gain stages hard-clip for a fizzy, gritty bite. Mode 0 LP / 1 BP / 2 HP.

Param	Range	Default	Unit
Cutoff	20 – 18000	1000	Hz
Reso	0 – 1	0.4	—
Mod	0 – 4	0	oct
Mode	LP · BP · HP		—

Acido 3 inputs

TB-303-style 4-pole diode ladder: high-resonance acid squelch with a single diode-shaped feedback stage.

Param	Range	Default	Unit
Cutoff	20 – 18000	800	Hz
Reso	0 – 1	0.6	—
Mod	0 – 4	0	oct
Drive	1 – 8	1.5	—

Pultec 3 inputs

Pultec EQP-1A-style passive EQ: low boost and cut use offset corners so they do not cancel (the boost-with-dip trick), plus gentle tube warmth.

Param	Range	Default	Unit
Low Boost	0 – 14	0	dB
Low Cut	0 – 17	0	dB
Low Freq	20 – 150	60	Hz
High Boost	0 – 18	0	dB
High Freq	3000 – 16000	5000	Hz
High Cut	0 – 16	0	dB

API 550 3 inputs

API 550-style 3-band EQ with proportional-Q (bands widen at low boost, tighten at high boost) and op-amp soft-clip bite.

Param	Range	Default	Unit
Low	-12 – 12	0	dB
Low Freq	30 – 400	100	Hz
Mid	-12 – 12	0	dB
Mid Freq	200 – 5000	800	Hz
High	-12 – 12	0	dB
High Freq	2500 – 20000	8000	Hz

NE 1073 3 inputs

Neve 1073-style EQ: fixed 12 kHz HF shelf, sweepable mid bell and LF shelf, with transformer low-frequency harmonic warmth.

Param	Range	Default	Unit
Low	-16 – 16	0	dB
Low Freq	35 – 220	60	Hz
Mid	-18 – 18	0	dB
Mid Freq	360 – 7200	1600	Hz
High	-16 – 16	0	dB

SSL E 3 inputs

SSL E-series-style 4-band parametric EQ: constant-Q (bandwidth stays put as you boost or cut) for a clean, surgical response.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	30 – 450	80	Hz
LMF	-15 – 15	0	dB
LMF Freq	200 – 2500	500	Hz
HMF	-15 – 15	0	dB
HMF Freq	600 – 7000	3000	Hz
High	-15 – 15	0	dB

SSL G 3 inputs

SSL G-series-style 4-band parametric EQ: proportional-Q (the more you boost or cut, the narrower the band) for a musical, self-limiting curve.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	30 – 450	80	Hz
LMF	-15 – 15	0	dB
LMF Freq	200 – 2500	500	Hz
HMF	-15 – 15	0	dB
HMF Freq	600 – 7000	3000	Hz
High	-15 – 15	0	dB

Manley 3 inputs

Manley Massive Passive-style EQ: wide bandwidth limits the available gain (opposite of proportional-Q), passive LC curves with tube even-harmonic warmth.

Param	Range	Default	Unit
Low	-11 – 11	0	dB
Low Freq	22 – 1000	100	Hz
Bandwidth	0 – 1	0.5	—
High	-11 – 11	0	dB
High Freq	560 – 20000	8000	Hz

BBE 3 inputs

BBE Sonic Maximizer-style processor: lifts the highs and lows for added clarity and punch.

Param	Range	Default	Unit
Process	0 – 1	0.4	—
LoContour	0 – 1	0.3	—
Mix	0 – 1	1	—

Helios 3 inputs

Helios Type 69-style console EQ: a low bell, a switchable peak-or-trough mid at stepped frequencies, and a 10 kHz high shelf - punchy and musical.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	60 – 400	100	Hz
Mid	-10 – 15	0	dB
Mid Freq	700 – 6000	1400	Hz
High	-16 – 12	0	dB

GML 3 inputs

GML 8200-style parametric EQ: ultra-clean, transparent constant-Q bands with no saturation - the surgical reference where boost and cut cancel exactly.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	30 – 800	120	Hz
Mid	-15 – 15	0	dB
Mid Freq	120 – 8000	1000	Hz

Param	Range	Default	Unit
High	-15 – 15	0	dB
High Freq	1000 – 20000	8000	Hz

Trident 3 inputs

Trident A-Range-style console EQ: four fixed musical bands (LF shelf, two mid bells, HF shelf) with an aggressive, forward voicing.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Lo Mid	-15 – 15	0	dB
Hi Mid	-15 – 15	0	dB
High	-15 – 15	0	dB

Chandler 3 inputs

Chandler Curve Bender-style EQ (EMI/Abbey Road): musical stepped bands with transformer/valve harmonic colour.

Param	Range	Default	Unit
Low	-15 – 15	0	dB
Low Freq	30 – 300	60	Hz
Mid	-15 – 15	0	dB
Mid Freq	200 – 6000	1000	Hz
High	-15 – 15	0	dB
High Freq	3000 – 20000	10000	Hz

API 560 3 inputs

API 560-style graphic EQ: fixed octave bands with proportional-Q so adjacent bands sum smoothly (4-band).

Param	Range	Default	Unit
31 Hz	-12 – 12	0	dB
125 Hz	-12 – 12	0	dB
1 kHz	-12 – 12	0	dB
8 kHz	-12 – 12	0	dB

MEQ5 3 inputs

Pultec MEQ-5-style midrange passive EQ: a low-mid boost, a mid dip and a high-mid boost for vocal/guitar presence.

Param	Range	Default	Unit
Low Mid	0 – 12	0	dB
Dip	0 – 12	0	dB
Hi Mid	0 – 12	0	dB

Maag 3 inputs

Maag EQ4-style EQ: musical low and mid bands plus the famous Air Band - a very high shelf for open, silky top.

Param	Range	Default	Unit
Low	-10 – 10	0	dB
Mid	-10 – 10	0	dB
Mid Freq	200 – 5000	1000	Hz
Air	0 – 10	0	dB

Avalon 3 inputs

Avalon VT-737-style tube channel EQ: clean, musical bell and shelf bands with gentle tube warmth.

Param	Range	Default	Unit
Low	-14 – 14	0	dB
Low Freq	30 – 300	80	Hz
Mid	-14 – 14	0	dB
Mid Freq	200 – 6000	1000	Hz
High	-14 – 14	0	dB
High Freq	3000 – 20000	10000	Hz

Filter 3 inputs

Multimode state-variable (TPT) filter switchable between low-pass, high-pass and band-pass via Mode. Cutoff tracks an exponential mod input (in 3) scaled by Mod in octaves, while Reso sets the Q up to self-oscillation and an optional pre-Drive (tanh) dirties the resonance. Mix blends the filtered result back against the dry signal.

Param	Range	Default	Unit
Cutoff	20 – 20000	1000	Hz
Reso	0.1 – 10	0.7	Q
Mode	0 – 2	0	—
Mod	0 – 4	0	oct
Drive	0 – 24	0	dB
Mix	0 – 1	1	—

AutoWah 1 input

Envelope-following resonant band-pass (auto-wah): the input level sweeps a band-pass from Base upward by Range, with Sens setting how strongly playing dynamics open it and Attack the follow speed. Reso sets the peak sharpness and Mix the blend. Funky on guitar, clav and bass; raise Reso for a vocal quack.

Param	Range	Default	Unit
Sens	0 – 1	0.6	—
Base	100 – 2000	400	Hz
Range	200 – 8000	3000	Hz
Reso	0.3 – 8	3	Q
Attack	1 – 200	15	ms
Mix	0 – 1	1	—

Formant 1 input

Morphing vowel filter: three parallel band-pass resonators tuned to formant frequencies are swept continuously across the A-E-I-O-U vowels by Vowel, with Q setting the formant sharpness and Mix the blend. Produces talking, vocal timbres. Sweep Vowel with an LFO or envelope for a wah-like vowel morph.

Param	Range	Default	Unit
Vowel	0 – 4	0	—
Q	2 – 20	10	—
Mix	0 – 1	1	—

EQ 1 input

Single parametric peaking EQ band (RBJ biquad): Gain boosts or cuts a bell centred at Freq, with Q setting its width - narrow for surgical notches, wide for broad tone shaping. Chain several for a full parametric EQ. See GraphicEQ for fixed bands and the analog pack (Pultec/Neve/SSL) for coloured curves.

Param	Range	Default	Unit
Freq	30 – 18000	1000	Hz
Gain	-18 – 18	0	dB
Q	0.3 – 8	1	—

Ladder 3 inputs

Moog-style 4-pole transistor-ladder low-pass: four cascaded one-poles with a tanh input Drive and global resonance feedback that self-oscillates near maximum Reso. Cutoff tracks an exponential mod input (in 3) in octaves. Warmer and more characterful than the clean Filter - push Drive into the ladder for growl; see the Mini analog model for a Minimoog-voiced variant.

Param	Range	Default	Unit
Cutoff	20 – 18000	1200	Hz
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Drive	1 – 8	1	—

Shelf 1 input

Low- or high-shelf EQ (RBJ shelving biquad; Type selects low or high): boosts or cuts everything below/above Freq by Gain at a fixed slope. Use for broad tonal tilt - warming the lows or adding air to the highs. Pair two for a tilt curve, or see TiltEQ for a single-knob version.

Param	Range	Default	Unit
Freq	30 – 18000	2000	Hz
Gain	-18 – 18	0	dB
Type	0 – 1	0	—

DJFilter 3 inputs

One-knob DJ filter: with Cutoff at centre the signal is open, turning down sweeps in a low-pass and turning up sweeps in a high-pass, while Reso adds bite at the corner. The Mod input (in 3) offsets the knob for automation. The classic build-and-drop sweep on a single control.

Param	Range	Default	Unit
Cutoff	0 – 1	0.5	—
Reso	0.1 – 8	0.7	Q
Mod	-1 – 1	0	—

Notch 1 input

Band-reject (notch) filter that removes a narrow band around Freq, with Q setting its width and Mix the blend. Use it to kill hum, a resonance or a feedback frequency, or sweep it for phaser-like effects. Narrow Q for surgical removal, wider for broad scooping.

Param	Range	Default	Unit
Freq	30 – 18000	1000	Hz
Q	0.3 – 16	4	—
Mix	0 – 1	1	—

Wah 3 inputs

Manually-swept resonant band-pass: Pedal sweeps the peak across Range, with Reso setting the quack and Mix the blend. Drive Pedal from the mod input (in 3) - an LFO, envelope or mod wheel - for auto-wah, talk-box or pedal-wah effects. See AutoWah for a built-in envelope follower.

Param	Range	Default	Unit
Pedal	0 – 1	0.5	—
Range	200 – 3000	1800	Hz
Reso	1 – 10	5	Q
Mod	-1 – 1	0	—
Mix	0 – 1	1	—

TiltEQ 1 input

One-knob spectral tilt that splits the signal at a pivot frequency with a one-pole low-pass and rocks the low and high halves in opposite directions. Tilt sets the slope from -1 (low band up, high band down) through 0 (flat) to +1 (high band up, low band down), Freq places the pivot the split happens around, and Range scales the maximum boost/cut in dB applied to each half. Useful as a fast tone balancer for brightening or warming a source without reaching for a full EQ.

Param	Range	Default	Unit
Tilt	-1 – 1	0	—
Freq	100 – 8000	1000	Hz
Range	0 – 12	6	dB

Telephone 1 input

Band-limits the input to roughly 300 Hz-3.4 kHz by cascading a one-pole low-pass at 3.4 kHz into a one-pole high-pass at 300 Hz, leaving only the narrow midrange band that survives a phone line. Mix crossfades between the dry signal and the band-passed result. Use it for lo-fi telephone, intercom and AM-radio voice effects.

Param	Range	Default	Unit
Mix	0 – 1	1	—

SallenKey 3 inputs

Resonant low-pass built on a TPT state-variable core taking the low-pass output, with the cutoff frequency exponentially modulated by in 3 scaled by Mod in octaves. Cutoff sets the corner, Reso raises the Q toward self-oscillation, and Mix blends the filtered low-pass back against the dry input. The Sallen-Key topology gives a smooth, gentle resonance suited to warm synth and bass filtering.

Param	Range	Default	Unit
Cutoff	20 – 20000	1000	Hz

Param	Range	Default	Unit
Reso	0 – 1	0.3	—
Mod	0 – 4	0	oct
Mix	0 – 1	1	—

Steiner 1 input

Steiner-Parker style multimode filter whose TPT core derives low-pass, high-pass, band-pass and notch responses simultaneously, with Mode selecting which one is output. Cutoff sets the corner frequency, Reso controls the Q toward self-oscillation, and Mix crossfades the selected response against the dry signal. Switching modes covers everything from dark sweeps to hollow notch tones for synth and effect duty.

Param	Range	Default	Unit
Cutoff	20 – 20000	1000	Hz
Reso	0 – 1	0.3	—
Mode	0 – 3	0	—
Mix	0 – 1	1	—

Crossover 1 input

Two-band splitter that derives a low band from two cascaded one-pole low-passes (an LR2 response) and forms the high band by subtracting it from the input, outputting one band at a time. Freq sets the split frequency and Band chooses whether the low band or the high band is passed. Use it to route or isolate a frequency region, or to feed separate processing chains per band.

Param	Range	Default	Unit
Freq	50 – 5000	800	Hz
Band	0 – 1	0	—

Resonator 1 input

High-Q band-pass tuned to a musical pitch, converting the Pitch parameter in semitones to a centre frequency and ringing the input through it like a struck body. Pitch sets the resonant note, Reso sets the Q (sharpness and ring length), and Mix blends the doubled resonant output against the dry signal. Drive it with transients or noise for tuned percussion, pinged-body and comb-resonance textures.

Param	Range	Default	Unit
Pitch	24 – 108	60	st
Reso	1 – 40	12	Q
Mix	0 – 1	1	—

Fizz 1 input

Resonant TPT band-pass filter whose output is mixed with a rectified copy of itself, adding fizzy upper harmonics that grow with resonance. Cutoff sets the band centre (100-12000 Hz), Reso sets both the filter Q and the amount of rectified harmonic content, and Mix blends the result against the dry signal. Use it to add bright, buzzy edge and emphasis to a narrow frequency band.

Param	Range	Default	Unit
Cutoff	100 – 12000	2000	Hz
Reso	0 – 1	0.6	—
Mix	0 – 1	1	—

Rasp 1 input

Resonant TPT band-pass filter followed by an asymmetric tanh waveshaper that drives positive half-cycles harder than negative ones, producing a rasping, lopsided distortion. Cutoff sets the band centre (100-12000 Hz), Reso sets the Q and scales the drive so higher resonance increases the snarl, and Mix blends against the dry signal. Ranges from a soft growl at low settings to an aggressive scream at high resonance.

Param	Range	Default	Unit
Cutoff	100 – 12000	1500	Hz
Reso	0 – 1	0.6	—
Mix	0 – 1	1	—

Ripple 1 input

Tuned comb resonator built from a short delay line whose length tracks a musical pitch, with a low-passed feedback path that makes it ring at that frequency. Pitch sets the resonant note as a MIDI value (24-108 st), Reso sets the feedback amount up to 0.98 for longer, sharper ringing, and Mix blends the resonant tone against the dry signal. Pushes input toward a pitched, metallic ring keyed to the chosen note.

Param	Range	Default	Unit
Pitch	24 – 108	60	st
Reso	0 – 0.98	0.7	—
Mix	0 – 1	1	—

LPG 2 inputs

Vactrol-style low-pass gate: the control input (in 2) simultaneously opens a low-pass filter and a VCA, so brightness and level rise together as the control climbs. Response curves the control law from snappy to sluggish, emulating the vactrol's lag, and Mix blends against dry. Driven by an envelope or trigger it gives the plucky, organic dynamics of Buchla-style patches.

Param	Range	Default	Unit
Response	0 – 1	0.5	—
Mix	0 – 1	1	—

GraphicEQ 1 input

Three-band graphic EQ: one-pole filters at 250 Hz and 4 kHz split the input into low, mid and high bands, each scaled by its own +/-12 dB gain and summed back. Low, Mid and High set the per-band gain in dB and Mix blends against dry. A simple tonal shaping tool for broad low/mid/high balance rather than surgical correction.

Param	Range	Default	Unit
Low	-12 – 12	0	dB
Mid	-12 – 12	0	dB
High	-12 – 12	0	dB
Mix	0 – 1	1	—

Rings 1 input

Modal resonator bank: three TPT band-pass resonators tuned around a base pitch are excited by the input, summed to produce a struck/plucked resonant body. Pitch sets the fundamental in semitones, Structure morphs the partial spacing from harmonic (1, 2, 3) to inharmonic (1, 2.76, 5.40), Decay sets the resonator Q (ring length), and Bright scales the summed output level. Mix blends the resonant tone with the dry signal for physical-modeling and bell/string textures.

Param	Range	Default	Unit
Pitch	24 – 96	48	st
Structure	0 – 1	0.3	—
Decay	0 – 0.99	0.9	—
Bright	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Wobble 1 input

Tempo-coherent resonant low-pass wobble: an LFO phased from c.time sweeps a TPT low-pass cutoff between 150 Hz and roughly 6 kHz for dubstep-style movement. Rate sets the LFO frequency, Depth scales the cutoff sweep range, Reso sets the filter resonance, and Shape selects sine, ramp or square LFO contour. Mix blends the wobbled output with the dry signal.

Param	Range	Default	Unit
Rate	0.25 – 8	2	Hz
Depth	0 – 1	0.8	—
Reso	0.5 – 15	8	—
Shape	0 – 2	0	—
Mix	0 – 1	1	—

Sweep 1 input

Onset-triggered filter sweep: an amplitude envelope detects input onsets and restarts a ramp that drives a TPT low-pass cutoff from 100 Hz upward for riser/faller effects. Time sets the ramp duration, Range scales how far the cutoff travels, Reso sets the filter resonance, and Dir flips the ramp between rising and falling. Mix blends the swept output with the dry signal.

Param	Range	Default	Unit
Time	0.1 – 5	1	s
Range	0 – 1	1	—
Reso	0.5 – 15	3	—
Dir	0 – 1	0	—
Mix	0 – 1	1	—

Distortion

50 modules

Array 1 input

Drawable waveshaper: maps the input at in0 (folded from bipolar into a 0 to 1 index) through a transfer curve drawn in node config via linear interpolation, then back to bipolar, blended by Mix against the dry input. The drawn breakpoints define the input-to-output mapping and Mix sets the wet/dry balance. A diagonal curve is a clean pass-through while steeper or folded curves act as custom distortion and waveshaping.

Param	Range	Default	Unit
Mix	0 – 1	1	—

Mulholland 1 input

Tanh overdrive feeding a resonant Chamberlin state-variable lowpass, with a slow internal LFO wobbling the cutoff. Drive sets the input gain into the saturator (1..30), Cutoff sets the filter frequency (~80 Hz..14 kHz), Reso sets the filter Q, Mod sets how much the LFO modulates cutoff, and Mix blends wet against dry. Goes from gentle filtered grit to a screaming self-resonant squeal as Drive and Reso climb.

Param	Range	Default	Unit
Drive	1 – 30	8	—
Cutoff	0 – 1	0.5	—
Reso	0.5 – 12	6	—
Mod	0 – 1	0.2	—
Mix	0 – 1	1	—

NevePre 3 inputs

Neve 1073-style preamp: class-A with transformer low-frequency saturation - the bass 'sings' with odd-harmonic iron warmth.

Param	Range	Default	Unit
Drive	1 – 12	2	—
Bias	0 – 0.3	0.08	—
Tone	2000 – 18000	12000	Hz
Mix	0 – 1	1	—

ApiPre 3 inputs

API 312-style preamp: discrete op-amp with a harder knee and a bright top; the harmonic tilt shifts upward as level rises.

Param	Range	Default	Unit
Drive	1 – 12	2	—
Bias	0 – 0.2	0.04	—
Tone	3000 – 16000	8000	Hz
Mix	0 – 1	1	—

TubePre 3 inputs

Vintage tube console-style preamp: soft asymmetric triode clipping, 2nd-harmonic dominant, with a warm low shelf.

Param	Range	Default	Unit
Drive	1 – 16	3	—
Bias	0 – 0.4	0.15	—
Tone	0 – 6	2	dB
Mix	0 – 1	1	—

BusSat 3 inputs

SSL bus-style VCA saturation: a thin even-harmonic 'glue' veneer for the mix bus - subtle thickening, not obvious distortion.

Param	Range	Default	Unit
Drive	0 – 1	0.3	—
Bias	0 – 0.2	0.06	—
Mix	0 – 1	0.25	—

Studer 3 inputs

Studer A800-style tape: 3rd-harmonic soft saturation, a firm low-frequency head bump and gentle HF loss - punchy track-level glue.

Param	Range	Default	Unit
Drive	1 – 8	2	—
Head Bump	0 – 8	4	dB
HF Loss	4000 – 18000	12000	Hz
Mix	0 – 1	1	—

Ampex 3 inputs

Ampex ATR-102-style mastering tape: gentler saturation, a soft wide head bump and smooth HF roll-off - subtle mix-bus polish.

Param	Range	Default	Unit
Drive	1 – 6	1.5	—
Head Bump	0 – 6	2	dB
HF Loss	6000 – 18000	14000	Hz
Mix	0 – 1	0.7	—

TubeScreamer 3 inputs

Ibanez Tube Screamer-style overdrive: soft symmetric clipping in the feedback path with the famous ~720 Hz mid hump.

Param	Range	Default	Unit
Drive	1 – 50	15	—
Tone	1000 – 5000	2500	Hz
Level	0 – 1	0.5	—
Asym	0 – 0.3	0	—

RAT 3 inputs

ProCo RAT-style distortion: aggressive hard clipping to ground, tamed by a single low-pass Filter knob.

Param	Range	Default	Unit
Drive	1 – 80	20	—
Filter	500 – 6000	3000	Hz
Level	0 – 1	0.4	—

BigMuff 3 inputs

Big Muff Pi-style fuzz: two cascaded soft-clip stages that square up into a wall of sustain, with a scooped-mid tone.

Param	Range	Default	Unit
Sustain	1 – 40	15	—
Tone	0 – 1	0.5	—
Level	0 – 1	0.5	—

FuzzFace 3 inputs

Fuzz Face-style germanium fuzz: asymmetric clipping that cleans up as you lower the Input; gated, spitty character.

Param	Range	Default	Unit
Fuzz	1 – 60	20	—
Bias	0.4 – 0.9	0.65	—
Input	0 – 1	1	—
Level	0 – 1	0.5	—

Klon 3 inputs

Klon Centaur-style overdrive: germanium hard clipping summed back with the clean signal - 'transparent' via the dual-path blend, not soft clipping.

Param	Range	Default	Unit
Gain	1 – 30	8	—
Treble	0 – 6	2	dB
Blend	0 – 1	0.6	—
Level	0 – 1	0.6	—

DS1 3 inputs

Boss DS-1-style distortion: hard symmetric silicon clipping with an active tilt-tone control - brash and cutting.

Param	Range	Default	Unit
Dist	1 – 60	20	—
Tone	0 – 1	0.5	—
Level	0 – 1	0.4	—

MXRDist 3 inputs

MXR Distortion+-style overdrive: asymmetric germanium clipping in the feedback loop - smoother and woolier than a DS-1, with an output low-pass.

Param	Range	Default	Unit
Dist	1 – 50	15	—
Level	0 – 1	0.5	—

Octavia 3 inputs

Octavia-style octave-up fuzz: full-wave rectification doubles the pitch, then heavy fuzz - the ring-mod-like upper-octave scream.

Param	Range	Default	Unit
Drive	1 – 40	15	—
Tone	1000 – 8000	3500	Hz
Level	0 – 1	0.5	—

Rangemaster 3 inputs

Dallas Rangemaster-style treble booster: a high-pass into a germanium gain stage adds bright, singing harmonics on top of the dry signal.

Param	Range	Default	Unit
Boost	1 – 20	6	—
Range	200 – 2000	700	Hz
Level	0 – 1	0.6	—

Aphex 3 inputs

Aphex Aural Exciter-style enhancer: generates new high-frequency harmonics from a high-passed, distorted side path and blends them back for air and presence.

Param	Range	Default	Unit
Tune	1000 – 8000	3000	Hz
Harmonics	0 – 1	0.4	—
Mix	0 – 1	0.5	—

SuperFuzz 3 inputs

Univox Super-Fuzz-style octave fuzz: full-wave rectification for the octave-up into hard germanium square clipping, with a switchable 1 kHz mid-scoop notch.

Param	Range	Default	Unit
Drive	1 – 40	18	—
Tone	Flat · Scoop	—	—
Level	0 – 1	0.5	—

OCD 3 inputs

Fulltone OCD-style overdrive: asymmetric MOSFET hard clipping (smooth-edged) with a bright/dark voicing switch.

Param	Range	Default	Unit
Drive	1 – 40	12	—
Tone	1500 – 9000	4000	Hz
Voicing	LP · HP		—
Level	0 – 1	0.5	—

MetalZone 3 inputs

Boss MT-2 Metal Zone-style high-gain distortion: two clip stages into an active dual-peak mid scoop with a sweepable parametric mid.

Param	Range	Default	Unit
Gain	1 – 60	30	—
Mid Freq	240 – 5000	800	Hz
Mid Level	-15 – 15	0	dB
Scoop	0 – 1	0.6	—
Level	0 – 1	0.4	—

HeliosPre 3 inputs

Helios console-style preamp: punchy class-A transistor saturation with an asymmetric Bias and a forward midrange push (a fixed ~1.5 kHz bell) for presence. Drive sets the grit, Tone rolls the top and Mix blends dry/wet. Aggressive and upfront - the British rock console colour; see N1073/NevePre for a smoother transformer voicing.

Param	Range	Default	Unit
Drive	1 – 12	2	—
Bias	0 – 0.2	0.05	—
Tone	2000 – 16000	9000	Hz
Mix	0 – 1	1	—

V76 3 inputs

Telefunken V76-style tube preamp: rich, warm 2nd-harmonic tube saturation with a vintage low-mid body.

Param	Range	Default	Unit
Drive	1 – 16	3	—
Bias	0 – 0.4	0.18	—
Body	0 – 6	2	dB
Mix	0 – 1	1	—

ChandlerPre 3 inputs

Chandler TG2-style preamp (EMI/Abbey Road): transformer-coupled, punchy with a distinctive forward presence.

Param	Range	Default	Unit
Drive	1 – 12	2	—
Bias	0 – 0.2	0.06	—
Tone	2000 – 16000	8000	Hz

Param	Range	Default	Unit
Mix	0 – 1	1	—

Drive 1 input

Waveshaping overdrive with three curves (Type: tanh soft-clip, sine fold, or hard clip) driven by up to 48 dB of gain plus a Bias offset for even-harmonic asymmetry. A post Tone low-pass tames fizz, Mix blends dry/wet, and Output trims the final level. Spans subtle warmth to aggressive distortion depending on Drive and curve.

Param	Range	Default	Unit
Drive	0 – 48	12	dB
Output	-24 – 6	-6	dB
Bias	-1 – 1	0	—
Tone	200 – 18000	18000	Hz
Mix	0 – 1	1	—
Type	0 – 2	0	—

Wavefolder 1 input

West-coast wavefolder that multiplies the signal by Fold and reflects it through triangle folding boundaries, adding bright harmonics that bloom as level rises. Bias shifts the fold symmetry for asymmetric timbres, Mix blends dry/wet and Output trims gain. Unlike clipping, folding keeps energy moving, so the tone stays lively even at extreme settings.

Param	Range	Default	Unit
Fold	1 – 12	2	—
Bias	-1 – 1	0	—
Mix	0 – 1	1	—
Output	-24 – 6	0	dB

Bitcrush 1 input

Digital degradation: Bits quantizes the amplitude to 2^{Bits} levels (quantization noise) while Downsample holds each sample for N frames (sample-rate reduction and aliasing). Mix blends against the dry signal. Low Bits gives gritty lo-fi crunch; high Downsample adds metallic aliasing for retro and 8-bit textures.

Param	Range	Default	Unit
Bits	1 – 16	8	—
Downsample	1 – 64	4	x
Mix	0 – 1	1	—

Waveshaper 1 input

Memoryless waveshaper with four transfer curves (Shape: tanh, atan, cubic soft-clip, sine fold) driven by Drive and blended via Mix. Each curve adds a different harmonic signature, from smooth saturation (tanh/atan) to a foldback character (sine). Lighter-weight than Drive when you just want one fixed shaping curve.

Param	Range	Default	Unit
Drive	1 – 20	2	—
Shape	0 – 3	0	—
Mix	0 – 1	1	—

Exciter 1 input

Harmonic exciter: splits off the band above Freq, generates new upper harmonics from it with a tanh Drive, and blends them back via Amount for added air and presence. Subtle settings add sheen and intelligibility without raising broadband level. Most effective on dull vocals, drums and full mixes.

Param	Range	Default	Unit
Freq	500 – 12000	3000	Hz
Drive	1 – 20	4	—
Amount	0 – 1	0.4	—

Lofi 1 input

Lo-fi degrader: sample-rate reduction (holds each sample for Rate frames) plus a Tone low-pass and added tape Hiss, blended via Mix. Unlike Bitcrush it quantizes in time rather than bit-depth, layering filtering and noise for a warmer vintage-sampler grime. Push Rate for aliasing; lower Tone and raise Hiss for cassette/SP-style character.

Param	Range	Default	Unit
Rate	1 – 64	8	x
Tone	500 – 12000	4000	Hz
Hiss	0 – 1	0.05	—
Mix	0 – 1	1	—

Tube 1 input

Tube-style saturation: an asymmetric transfer curve (offset by Bias) adds the even and odd harmonics of a valve stage, tamed by a Tone low-pass and blended via Mix. Warmer and rounder than the harder Drive/Waveshaper - subtle settings thicken and glue, higher Drive gives singing tube grit. See the TubePre and VariMu analog models for circuit-specific versions.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Bias	0 – 1	0.3	—
Tone	500 – 16000	8000	Hz
Mix	0 – 1	1	—

Clip 1 input

Clipper that morphs from soft (tanh) to hard clipping at the Threshold via Hardness, with Output trim and dry/wet Mix. Soft adds rounded saturation harmonics; hard adds bright odd harmonics and caps peaks dead. Use as a tone shaper or a transparent peak ceiling ahead of a limiter.

Param	Range	Default	Unit
Threshold	-24 – 0	-6	dB
Hardness	0 – 1	0.5	—
Output	-12 – 12	0	dB
Mix	0 – 1	1	—

Shaper 1 input

SynthMaster-style 12-mode waveshaper: Shape selects soft/hard clip, tube, diode, sine and triangle fold, full/half rectify, sine, crush, sync-wrap and Chebyshev modes, each driven by Drive with Bias for symmetry. Mix and Output blend and trim. Modulate Shape to morph between distortion characters - a one-stop tone-shaping distortion.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Shape	0 – 11	0	—
Bias	-1 – 1	0	—
Mix	0 – 1	1	—
Output	-24 – 6	0	dB

Rectify 1 input

Rectifier waveshaper: full-wave (Mode 0) folds the negative half upward to add strong octave-up and even harmonics, while half-wave (Mode 1) keeps only the positive half. Mix blends with the dry signal. The classic octave-fuzz colour - aggressive alone, useful in small amounts for harmonic lift.

Param	Range	Default	Unit
Mode	0 – 1	0	—
Mix	0 – 1	1	—

Fuzz 1 input

High-gain fuzz: heavy drive soft-saturates then hard-clips into a square wave for a thick, sustaining buzz, with a Tone low-pass to tame the fuzz and Level to set output. Mix blends dry/wet. Aggressive and compressed - see the BigMuff and FuzzFace analog models for circuit-specific fuzz voicings.

Param	Range	Default	Unit
Drive	0 – 48	24	dB
Tone	0 – 1	0.5	—
Level	-24 – 6	-8	dB
Mix	0 – 1	1	—

OctaveFuzz 1 input

Octave fuzz: blends a fuzz-distorted signal with a DC-corrected full-wave-rectified upper octave, the classic Hendrix/Octavia scream. Drive sets gain, Octave the upper-octave blend and Mix dry/wet. Tracks best on clean single notes near the 12th fret; chords inter-modulate, by design.

Param	Range	Default	Unit
Drive	0 – 48	24	dB
Octave	0 – 1	0.6	—
Mix	0 – 1	1	—

Chebyshev 1 input

Chebyshev-polynomial shaper that synthesizes a single chosen Harmonic (2nd-8th) from the input via the T_n recurrence, blended by Amount and Mix. Unlike broadband clipping it adds one targeted overtone - the 2nd for octave thickness, higher orders for metallic ring. Pairs well with a clean sine input for precise harmonic generation.

Param	Range	Default	Unit
Harmonic	2 – 8	2	—
Amount	0 – 1	0.5	—
Mix	0 – 1	1	—

SubEnhance 1 input

Sub-bass enhancer: a flip-flop toggled per rising zero crossing generates a half-frequency square tracked by the input envelope and low-passed, added under the dry signal at Sub. Reinforces weak low end on kicks, bass and 808s. Monophonic by nature - best on clean low-register sources.

Param	Range	Default	Unit
Sub	0 – 1	0.5	—
Mix	0 – 1	1	—

TapeSat 1 input

Tape-style saturation: a soft tanh stage with a touch of linear blend emulates magnetic-tape compression, with Warmth rolling off the highs for a rounder tone and Mix blending dry/wet. Gentle glue and harmonic thickening rather than overt distortion. See the Studer and Ampex analog models for machine-specific tape.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Warmth	0 – 1	0.4	—
Mix	0 – 1	1	—

XformerSat 1 input

Transformer saturation: asymmetric soft-clipping knees (different for positive and negative swings) add even-harmonic colour, plus a low-passed component for iron-core low-end thickening. Subtle weight and warmth on bass, drums and the mix bus. See the NevePre analog model for a transformer-preamp variant.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Mix	0 – 1	1	—

DynSat 1 input

Dynamic saturation: the drive amount tracks the input envelope, so louder passages distort more and quiet ones stay clean (Sens sets how strongly). Adds harmonics on transients and peaks without muddying sustained low-level material. Good for program-dependent grit on drums and bus.

Param	Range	Default	Unit
Drive	0 – 36	6	dB
Sens	0 – 1	0.5	—
Mix	0 – 1	1	—

Erosion 1 input

Noise-modulation distortion: sprays band-limited noise (Freq sets its colour) onto the signal by multiplying it with the input, eroding the tone into gritty granular texture. Amount sets intensity and Mix dry/wet. From subtle dirt to harsh destruction - a creative degrade rather than a clean overdrive.

Param	Range	Default	Unit
Amount	0 – 1	0.4	—
Freq	0 – 1	0.5	—
Mix	0 – 1	1	—

Character 1 input

Character drive with five voicings (Mode): Push (soft clip), Heat (S-clip), Soar (sine fold), Howl (triangle fold) and Shred (wrap fold), each driven by Drive and blended via Mix. One module spanning warm saturation to wild folding distortion. Modulate Mode or Drive for evolving textures.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Mode	0 – 4	0	—
Mix	0 – 1	1	—

Vinyl 1 input

Vinyl character that reads the signal through a slowly modulated delay for wow pitch-drift, injects random impulses for surface crackle, and pushes the result through a tanh saturator for groove distortion. Wow scales the pitch-drift depth, Crackle sets how often surface pops appear, Drive sets the saturation amount, and Mix sets dry/wet. Use it to age a clean source with turntable wobble, dust noise and warm grit.

Param	Range	Default	Unit
Crackle	0 – 1	0.4	—
Wow	0 – 1	0.3	—
Drive	0 – 1	0.3	—
Mix	0 – 1	1	—

AmpSim 1 input

Guitar amp model chaining an asymmetric tanh preamp, a tone-tilt filter and a speaker-cabinet low-pass with output compensation. Drive sets preamp gain (0-48 dB) into the asymmetric clipper, Tone tilts between darker and brighter response, and Cab sets the cabinet cutoff (brighter at higher values). Mix blends the modeled tone against the dry; ranges from edge-of-breakup grit to saturated lead tones.

Param	Range	Default	Unit
Drive	0 – 48	18	dB
Tone	0 – 1	0.5	—
Cab	0 – 1	0.7	—
Mix	0 – 1	1	—

MultiDist 1 input

Three-band distortion: one-pole filters at 300 Hz and 3 kHz split the input into low, mid and high bands, each saturated through its own tanh stage before being summed. Low, Mid and High set the drive into each band's saturator and Mix blends the result against dry. Per-band drive lets you grind the highs without muddying the lows, or add weight to bass alone.

Param	Range	Default	Unit
Low	0 – 1	0.3	—
Mid	0 – 1	0.3	—
High	0 – 1	0.3	—
Mix	0 – 1	1	—

CabSim 1 input

Generic speaker-cabinet voicing: a 90 Hz high-pass trims sub rumble, a high-frequency rolloff tames fizz, and a peak around 1.5 kHz adds cabinet body, all blended by Mix. Body sets how much of the resonant midrange peak is added and Bright raises the

rolloff corner for a more open top end. A lightweight tone-shaping voicing for amp signals, not an impulse-response capture of a real cabinet.

Param	Range	Default	Unit
Body	0 – 1	0.5	—
Bright	0 – 1	0.5	—
Mix	0 – 1	1	—

Diode 1 input

Asymmetric diode clipper that drives the input into an exponential soft-knee saturation, shaping the positive and negative halves differently. Drive sets the input gain in dB into the nonlinearity, while Bias increases the curvature of the negative half for stronger asymmetry and added even harmonics. Mix blends the clipped signal against the dry input for germanium/silicon-style warmth and grit.

Param	Range	Default	Unit
Drive	0 – 36	12	dB
Bias	0 – 1	0.3	—
Mix	0 – 1	1	—

VHS 1 input

Cassette/VHS tape emulation: the input runs through a short modulated delay whose time wobbles via an internal 0.7 Hz LFO, then a one-pole low-pass rolls off highs and random pseudo-noise hiss is summed in. Wow sets the LFO delay-modulation depth (pitch wobble), Hiss scales the added noise level, and Dropout raises the probability of brief signal dropouts driven by a per-sample random envelope. Mix blends the degraded signal against the dry input for subtle aging or full lo-fi destruction.

Param	Range	Default	Unit
Wow	0 – 1	0.4	—
Hiss	0 – 1	0.3	—
Dropout	0 – 1	0.2	—
Mix	0 – 1	1	—

Subharmonic 1 input

Subharmonic generator: zero-crossing detection on the input drives a divide-by-four flip-flop, multiplying the input's rectified amplitude to synthesize a tone two octaves below, then smooths it with a one-pole low-pass. Sub sets the level of the generated sub, Tone opens the low-pass cutoff (darker to brighter sub), and Mix sums the sub onto the dry signal. Adds weight and low-end body to bass, kicks and leads.

Param	Range	Default	Unit
Sub	0 – 1	0.6	—
Tone	0 – 1	0.5	—
Mix	0 – 1	1	—

Amp

10 modules

Tweed 3 inputs

Fender Tweed 5E3-style amp: two soft-clip stages into a heavy power-amp sag that blooms under pick attack - warm, loose, early breakup.

Param	Range	Default	Unit
Gain	1 – 20	4	—
Tone	0 – 1	0.5	—
Sag	0 – 1	0.6	—
Level	0 – 1	0.5	—

Blackface 3 inputs

Fender Blackface-style amp: scooped-mid clean headroom (the ~500 Hz notch) with a stiff power section and a bright, glassy top.

Param	Range	Default	Unit
Gain	1 – 12	2	—
Treble	0 – 1	0.6	—
Bass	0 – 1	0.5	—
Bright	0 – 1	0.3	—
Level	0 – 1	0.6	—

Plexi 3 inputs

Marshall Plexi-style amp: mid-forward crunch (the raised ~700 Hz bump) with a presence shelf and aggressive output clipping.

Param	Range	Default	Unit
Gain	1 – 30	6	—
Mid	0 – 1	0.6	—
Presence	0 – 1	0.4	—
Master	0 – 1	0.6	—

AC30 3 inputs

Vox AC30 Top Boost-style amp: chimey top-boost EQ before a no-feedback class-A power stage, voiced by a sweepable Cut low-pass.

Param	Range	Default	Unit
Gain	1 – 20	5	—
Treble	0 – 1	0.6	—
Bass	0 – 1	0.4	—
Cut	0 – 1	0.3	—
Master	0 – 1	0.6	—

Recto 3 inputs

Mesa Rectifier-style high-gain amp: several cascaded saturation stages with tight inter-stage high-passing, a deep mid scoop and selectable power sag.

Param	Range	Default	Unit
Gain	1 – 40	15	—
Scoop	0 – 1	0.6	—
Presence	0 – 1	0.5	—
Sag	0 – 1	0.3	—
Master	0 – 1	0.5	—

Ampeg 3 inputs

Ampeg SVT-style bass amp: thick low-mid focus with gentle tube grind and a dark cab - built for bass.

Param	Range	Default	Unit
Gain	1 – 20	3	—
Bass	0 – 1	0.6	—
Mid	0 – 1	0.5	—
Master	0 – 1	0.6	—

Friedman 3 inputs

Friedman BE-100-style amp: hot-rodged Marshall high-gain - tight low end, singing mids, modern saturation.

Param	Range	Default	Unit
Gain	1 – 40	12	—
Mid	0 – 1	0.6	—
Presence	0 – 1	0.5	—
Master	0 – 1	0.5	—

Soldano 3 inputs

Soldano SLO-100-style amp: smooth, saturated high-gain lead tone with a singing sustain and a bright top.

Param	Range	Default	Unit
Gain	1 – 40	15	—
Tone	0 – 1	0.5	—
Presence	0 – 1	0.5	—
Master	0 – 1	0.5	—

Diezel 3 inputs

Diezel VH4-style amp: ultra-tight, aggressive German high-gain with a scooped, percussive low end.

Param	Range	Default	Unit
Gain	1 – 40	18	—
Scoop	0 – 1	0.6	—
Presence	0 – 1	0.5	—
Master	0 – 1	0.5	—

ENGL 3 inputs

ENGL-style amp: tight, scooped modern-metal high-gain with a focused attack and a dark cab.

Param	Range	Default	Unit
Gain	1 – 40	16	—
Scoop	0 – 1	0.7	—
Presence	0 – 1	0.5	—
Master	0 – 1	0.5	—

Modulation

43 modules

SwitcherLFO 0 inputs

Massive X morphing LFO: runs one phase at Rate through a bank of 8 bipolar shapes (sine, triangle, saw up/down, square, narrow pulse, s-curve, double sine), crossfading between adjacent shapes and mapping the result to a 0 to 1 output. Rate sets the cycle speed, Shape selects and morphs continuously across the bank, and Phase offsets the read position. Sweeping Shape gives a continuous waveform-morph modulator rather than a fixed LFO shape.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Shape	0 – 1	0	—
Phase	0 – 1	0	—

DAHDSR 1 input

Six-stage modulation envelope driven by the gate at in0: on gate-on it runs Delay, Attack, Hold, Decay then holds at Sustain, and on gate-off it enters Release, outputting a 0 to 1 level. Delay/Attack/Hold/Decay/Release are times in seconds and Sustain is the held level. The extra Delay and Hold stages over a plain ADSR let it shape slow swells or gated bursts as a modulation source.

Param	Range	Default	Unit
Delay	0 – 4	0	s
Attack	0 – 5	0.01	s
Hold	0 – 4	0	s
Decay	0 – 5	0.2	s
Sustain	0 – 1	0.7	—
Release	0 – 5	0.3	s

Glide 2 inputs

Portamento: one-pole smooths the pitch/control signal at in0 toward each new value over Time, gated by in1. Time sets the glide duration in seconds (zero snaps instantly), and Mode chooses Always (glide on every note change) or Legato (snap on an isolated note, glide only between overlapping notes). Drop it between a note source and an oscillator for per-layer portamento.

Param	Range	Default	Unit
Time	0 – 2	0.1	s
Mode	0 – 1	0	—

PathLFO 0 inputs

Drawable XY-path LFO: a Rate-driven phase traverses the loop of 2D waypoints set in node config and reads back one coordinate as a bipolar -1 to 1 output, scaled by Level. Rate sets the traversal speed, Axis chooses whether the X or Y coordinate drives the output, and Level scales the result. Pair two nodes set to X and Y on the same drawn path to modulate two targets in a coordinated 2D motion.

Param	Range	Default	Unit
Rate	0.01 – 50	1	Hz
Axis	0 – 1	0	—
Level	0 – 1	1	—

MSEG 1 input

Multi-segment envelope generator: a gate edge resets a phase that ramps from 0 to 1 over Rate seconds, and the output linearly interpolates between user-defined 'time,level' breakpoints parsed from the node config. Rate sets the total run time of the shape, Loop makes the phase wrap and repeat instead of holding the last level, and Level scales the final output. Breakpoints come from a 't,l;t,l;...' config string (defaulting to a fast attack and gentle decay when fewer than two are given), giving a flexible Serum/Vital-style modulation source for arbitrary multi-stage curves.

Param	Range	Default	Unit
Rate	0.05 – 8	1	s
Loop	0 – 1	0	—
Level	0 – 1	1	—

Barberpole 1 input

Infinite (barberpole) phaser: two 4-stage allpass banks a half-cycle apart, each amplitude-windowed so one fades in as the other wraps, giving an endless rising or falling sweep. Rate sets the sweep speed (Hz), Depth scales the allpass coefficient swing, Dir flips between rising and falling, and Mix blends wet against dry. A Shepard-tone style effect that seems to climb or descend forever.

Param	Range	Default	Unit
Rate	0.02 – 4	0.3	Hz
Depth	0 – 1	0.7	—
Dir	0 – 1	0	—
Mix	0 – 1	0.5	—

PeakPhaser 1 input

A 6-stage allpass phaser whose sweep is driven by a peak envelope follower on the input, so transients trigger the filter motion instead of an LFO. Soften sets the envelope release time, Feedback sets the allpass regeneration (up to 0.9), Amount scales how far each peak sweeps the phaser, and Mix blends wet against dry. Part of the Traktor Pulse family, it pumps the phaser in time with the signal's own dynamics.

Param	Range	Default	Unit
Soften	0 – 1	0.5	—
Feedback	0 – 0.9	0.5	—
Amount	0 – 1	0.7	—
Mix	0 – 1	0.5	—

TriceraChorus 1 input

Tri-voice chorus with a micro-detune swirl: three LFO-modulated delay taps at spread phases plus a slow pitch-detuned voice for a rich animated widening.

Param	Range	Default	Unit
Rate	0.05 – 6	0.5	Hz
Depth	0 – 1	0.6	—
Detune	0 – 1	0.3	—

Param	Range	Default	Unit
Mix	0 – 1	0.5	—

DimensionD 3 inputs

Roland Dimension D-style BBD chorus: two delay lines modulated in anti-phase for a wide, 'motionless' thickening with no audible wobble.

Param	Range	Default	Unit
Mode	1 · 2 · 3 · 4		—
Mix	0 – 1	0.5	—

CE1 3 inputs

Boss CE-1-style BBD chorus/vibrato: a triangle-LFO chorus or a wet-only vibrato, with preamp coloration.

Param	Range	Default	Unit
Mode	Chorus · Vibrato		—
Rate	0.1 – 8	1	Hz
Depth	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Solina 3 inputs

Solina/string-machine-style ensemble: three delay taps on 120-degree-phased dual LFOs (slow + fast) for a lush, moving multi-voice swirl.

Param	Range	Default	Unit
Slow	0.1 – 2	0.6	Hz
Fast	1 – 12	6	Hz
Depth	0 – 1	0.6	—
Mix	0 – 1	0.5	—

ElectricMistress 3 inputs

EHX Electric Mistress-style flanger: a swept short delay forms moving comb notches; Filter Matrix freezes the sweep for a fixed metallic tone.

Param	Range	Default	Unit
Rate	0.05 – 8	0.4	Hz
Range	0 – 1	0.5	—
Color	0 – 1	0.4	—
Matrix	Sweep · Freeze		—

TriStereo 3 inputs

Dytrronics Tri-Stereo Chorus-style ensemble: three BBD voices at staggered rates for a lush, wide three-dimensional chorus.

Param	Range	Default	Unit
Rate	0.1 – 3	0.5	Hz
Depth	0 – 1	0.6	—

Param	Range	Default	Unit
Mix	0 – 1	0.5	—

MXRFlanger 3 inputs

MXR Flanger-style BBD flanger: a deep, metallic swept comb with regeneration for jet-plane sweeps.

Param	Range	Default	Unit
Rate	0.05 – 6	0.3	Hz
Depth	0 – 1	0.7	—
Regen	0 – 0.9	0.4	—
Mix	0 – 1	0.5	—

Phase90 3 inputs

MXR Phase 90-style 4-stage phaser: four identical all-pass stages give two evenly-tracking notches; Script (clean) or Block (resonant feedback).

Param	Range	Default	Unit
Rate	0.05 – 5	0.5	Hz
Block	Script · Block	—	—
Mix	0 – 1	0.5	—

SmallStone 3 inputs

EHX Small Stone-style 4-stage OTA phaser: the Color switch adds feedback and widens the notch sweep for a more vocal, resonant swirl.

Param	Range	Default	Unit
Rate	0.05 – 5	0.4	Hz
Color	Off · On	—	—
Mix	0 – 1	0.5	—

UniVibe 3 inputs

Uni-Vibe-style 4-stage phaser: staggered (unevenly spaced) all-pass frequencies driven by an asymmetric photocell/lamp LFO - the watery, throbbing swirl. Chorus or Vibrato.

Param	Range	Default	Unit
Rate	0.05 – 8	2	Hz
Depth	0 – 1	0.7	—
Mode	Chorus · Vibrato	—	—
Mix	0 – 1	0.5	—

OptoTrem 3 inputs

Fender optical tremolo: a photocell amplitude modulation with a hard, choppy near-square throb.

Param	Range	Default	Unit
Rate	0.5 – 12	5	Hz

Param	Range	Default	Unit
Depth	0 – 1	0.6	—
Shape	0 – 1	0.7	—

BiasTrem 3 inputs

Tube bias tremolo: a smooth wavering amplitude modulation that also shifts harmonic content (cold-bias odd harmonics) for a richer throb.

Param	Range	Default	Unit
Rate	0.5 – 10	4	Hz
Depth	0 – 1	0.5	—
Bias	0 – 1	0.3	—

FuncGen 0 inputs

H3000-style function generator: a modulation source with 19 selectable waveshapes plus random sample-and-hold, smooth random and noise (no audio input).

Param	Range	Default
Rate	0.01 – 30	1
Shape	Sine · Triangle · Saw Up · Saw Down · Square · Pulse · Sine2 · Rectified · Half Sine · Exp Up · Exp Down · Log · Trapezoid · Staircase · Random	
Depth	0 – 1	1
Phase	0 – 1	0
Width	0.05 – 0.95	0.5
Bias	-1 – 1	0

Random 0 inputs

Self-contained random modulation source: latches a new random value at Rate (internal sample-and-hold of a noise source) with Smooth slewing between steps - from hard random steps to smooth wandering drift. Depth scales the swing and Bias offsets the centre. Needs no input; patch its output into any mod input for evolving, never-repeating movement.

Param	Range	Default	Unit
Rate	0.01 – 50	2	Hz
Smooth	0 – 1	0	—
Depth	0 – 1	1	—
Bias	-1 – 1	0	—

LF0 0 inputs

Low-frequency control generator with sine, triangle, saw and pulse shapes (Shape), free-running at Rate. Depth scales the swing, Bias offsets the centre, Phase shifts the start point, and Width sets the pulse duty cycle. Patch its output into any parameter's mod input for vibrato, tremolo, filter sweeps and rhythmic movement.

Param	Range	Default	Unit
Rate	0.01 – 20	1	Hz
Shape	0 – 3	0	—
Depth	0 – 1	1	—
Phase	0 – 1	0	—
Bias	-1 – 1	0	—
Width	0.05 – 0.95	0.5	—

RingMod 3 inputs

Ring and amplitude modulator that multiplies the input by a carrier on in 2 (plus Bias). Mode morphs from true ring modulation (bipolar carrier, inharmonic metallic sum/difference tones) to amplitude modulation (unipolar, keeps the original plus sidebands), and Amount sets dry/wet. An audio-rate carrier yields clangorous bell tones; a low-rate carrier becomes tremolo.

Param	Range	Default	Unit
Amount	0 – 1	1	—
Mode	0 – 1	0	—
Bias	-1 – 1	0	—

SampleHold 1 input

Clocked sample-and-hold that latches the input value at Rate and holds it until the next tick, producing stepped random or quantized control. Glide slews between steps for a smoother, portamento-like contour. Feed it noise for classic random sample-and-hold modulation, or an audio source to quantize it in time.

Param	Range	Default	Unit
Rate	0.1 – 200	8	Hz
Glide	0 – 1	0	—

Slew 1 input

Slew-rate limiter that caps how fast the signal can rise and fall, with independent Rise and Fall rates. Use it as portamento/glide on pitch CV, a smoother on stepped modulation, or a lag to de-zipper control changes. Asymmetric Rise/Fall gives simple envelope-like attack and decay shaping.

Param	Range	Default	Unit
Rise	0 – 1	0.5	—
Fall	0 – 1	0.5	—

Chaos 0 inputs

Chaos modulator driven by the logistic map ($x = r * x * (1 - x)$): at each Rate tick it iterates the map, and Chaos raises r from the periodic into the chaotic regime for unpredictable-but-deterministic motion. Smooth slews between values for a flowing control source rather than hard steps. Use it for evolving, never-quite-repeating modulation.

Param	Range	Default	Unit
Rate	0.1 – 500	20	Hz
Chaos	0 – 1	0.8	—
Smooth	0 – 1	0.5	—

Phaser 1 input

All-pass cascade phaser: an LFO sweeps 2-8 all-pass stages (Stages) to move a series of notches through the spectrum, with Feedback sharpening their resonance. Depth sets the sweep range and Mix the dry/wet blend. More stages add more notches for a deeper, more vocal swirl.

Param	Range	Default	Unit
Rate	0.05 – 8	0.4	Hz
Depth	0 – 1	0.7	—
Feedback	0 – 0.9	0.3	—

Param	Range	Default	Unit
Stages	2 – 8	4	—
Mix	0 – 1	0.5	—

Tremolo 0 inputs

Amplitude tremolo: an LFO at Rate modulates the signal level by Depth, switchable between a smooth sine and a choppy square (Shape). Classic for rhythmic volume pulsing, surf-guitar wobble and rotary-style throb. At full Depth with the square shape it becomes a hard on/off gate.

Param	Range	Default	Unit
Rate	0.05 – 20	5	Hz
Depth	0 – 1	0.5	—
Shape	0 – 1	0	—

Dropout 1 input

Random rhythmic muting: a clock at Rate decides per cycle whether to drop the signal to silence (Amount sets the drop probability), smoothed by Smooth to avoid clicks. Emulates a bad cable, dying battery or stuttery transmission. Subtle for movement, extreme for glitch and IDM textures.

Param	Range	Default	Unit
Rate	0.1 – 30	4	Hz
Amount	0 – 1	0.3	—
Smooth	0 – 1	0.5	—

TranceGate 1 input

Rhythmic hard gate: an LFO at Rate chops the signal on and off with a Duty cycle, smoothed by Smooth and scaled by Depth. Modulate Rate to lock it to tempo for the classic trance/EDM stutter. See StepSeq for arbitrary per-step patterns rather than a fixed duty.

Param	Range	Default	Unit
Rate	0.1 – 30	8	Hz
Duty	0.05 – 0.95	0.5	—
Smooth	0 – 1	0.2	—
Depth	0 – 1	1	—

Vector 4 inputs

Bilinear XY vector mixer: blends four inputs across an X/Y plane (in 1/in 2 the top corners, in 3/in 4 the bottom), with X and Y setting the morph position. Classic vector synthesis (Prophet VS / Wavestation) - feed it four oscillators or sources and automate X/Y to morph between them.

Param	Range	Default	Unit
X	0 – 1	0.5	—
Y	0 – 1	0.5	—

StepSeq 0 inputs

Step sequencer modulation source: cycles through up to 6 step levels (S1-S6) at Rate steps per second, outputting the held value of the current step. Use it as a rhythmic mod source for cutoff, pitch or level, or to chop gain into patterns. Modulate Rate to lock to tempo; see TranceGate for a simple on/off duty gate.

Param	Range	Default	Unit
Rate	0.1 – 30	4	Hz
Steps	1 – 6	4	—
S1	0 – 1	1	—
S2	0 – 1	0.5	—
S3	0 – 1	0.75	—
S4	0 – 1	0.25	—
S5	0 – 1	0.6	—
S6	0 – 1	0.4	—

Drift 0 inputs

Slow smoothed random-walk modulator centred on 0.5, emulating vintage analog drift. Rate sets how often it picks a new target and Depth the deviation. Patch a little into pitch, cutoff or level to take the static, too-perfect edge off a patch.

Param	Range	Default	Unit
Rate	0.05 – 5	0.5	Hz
Depth	0 – 1	0.5	—

Allpass 1 input

All-pass diffuser: a cascade of 1-8 first-order all-pass sections (Stages) that rotate phase while leaving magnitude flat, set by Coef. A building block for reverb diffusion, phasers and stereo de-correlation rather than an audible filter on its own. More stages and higher Coef smear transients harder.

Param	Range	Default	Unit
Coef	-0.95 – 0.95	0.7	—
Stages	1 – 8	4	—
Mix	0 – 1	1	—

LorenzLFO 0 inputs

Lorenz strange-attractor chaos modulator (Serum-style): integrates the Lorenz system and outputs one Axis (X/Y/Z), normalized and scaled by Depth. Produces smooth, organic, never-repeating motion - more flowing than stepped random. Great for evolving cutoff, pitch or pan drift.

Param	Range	Default	Unit
Rate	0.01 – 50	2	Hz
Axis	0 – 2	0	—
Depth	0 – 1	1	—

Combine 2 inputs

Combines two modulation or audio inputs (Pigments-style): Mode selects sum, ring (multiply), max, min, difference or sign-XOR, scaled by Mix. A math/logic node for the mod matrix - build complex modulators from simpler ones, or use it as a flexible ring/AM stage. Complements the Expr script node for inline math.

Param	Range	Default	Unit
Mode	0 – 5	0	—
Mix	0 – 1	1	—

TZFlanger 1 input

Through-zero flanger that reads two short delay taps swept in opposite directions by a cosine LFO, so the offsets cross and produce a deep tape-style null. Rate sets the LFO speed, Depth scales the sweep range, Feedback (up to 0.95) regenerates one tap for sharper resonance, and Mix sets dry/wet. The counter-swept design gives the dramatic zero-crossing whoosh that fixed flangers cannot reach.

Param	Range	Default	Unit
Rate	0.01 – 8	0.3	Hz
Depth	0 – 1	0.6	—
Feedback	0 – 0.95	0.4	—
Mix	0 – 1	0.5	—

Vibe 1 input

Photocell-style vibe that runs the signal through four staggered first-order all-pass stages whose coefficients are swept by a cosine LFO, each stage offset slightly to mimic mismatched optical cells. Rate sets the swirl speed, Depth scales how far the all-pass coefficients move, and Mix blends the phase-shifted output against the dry signal. The uneven staging gives the watery, throbbing swirl associated with classic vibe pedals.

Param	Range	Default	Unit
Rate	0.05 – 8	1	Hz
Depth	0 – 1	0.7	—
Mix	0 – 1	0.5	—

HarmTremolo 1 input

Harmonic tremolo that splits the signal into low and high bands with a one-pole low-pass and amplitude-modulates them in opposition with a sine LFO, so as the lows swell the highs dip and back again. Rate sets the LFO speed, Depth sets the modulation intensity, Freq places the band split, and Mix sets dry/wet. The opposing band motion gives a richer, more vocal sweep than amplitude tremolo.

Param	Range	Default	Unit
Rate	0.1 – 12	4	Hz
Depth	0 – 1	0.6	—
Freq	200 – 3000	800	Hz
Mix	0 – 1	1	—

WowFlutter 1 input

Tape pitch instability emulated by a short delay line (centred near 5 ms) whose read time is modulated by two sine LFOs, a slow ~0.6 Hz wow and a fast ~7 Hz flutter, producing continuous pitch drift. Wow scales the slow drift depth and Flutter scales the fast wobble, while Mix blends the warped signal against the dry input. Use low amounts for subtle tape warmth and higher settings for seasick, detuned movement.

Param	Range	Default	Unit
Wow	0 – 1	0.4	—
Flutter	0 – 1	0.3	—
Mix	0 – 1	1	—

Param	Range	Default	Unit
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Dimension 1 input

Dimension-D-style BBD chorus: a slow ~0.5 Hz LFO modulates a short delay around 8 ms, with the right channel running anti-phase, and the wet copy is blended with the dry at a fixed ratio. Depth scales how far the delay time is swept. Out picks L or R for the per-channel modulation phase, giving the lush stereo movement of the classic unit rather than a deep vibrato.

Param	Range	Default	Unit
Depth	0 – 1	0.5	—
Out	0 – 1	0	—

Gate 1 input

Step-sequenced tremolo gate that reads a per-step on/off pattern from a 16-bit mask, advancing through the steps on a sample-clock phase for tempo coherence. Rate sets the cycle frequency, Pattern is the bitmask of active steps, Steps sets how many steps are used, and Smooth softens the transitions between open and closed. Mix sets how deeply the gate cuts the signal, from subtle pulsing to full rhythmic chopping.

Param	Range	Default	Unit
Rate	0.25 – 8	2	Hz
Pattern	0 – 65535	43690	—
Steps	1 – 16	8	—
Smooth	0 – 1	0.3	—
Mix	0 – 1	1	—

Undulator 1 input

Modulated short delay combined with an envelope swell and a tremolo running at twice the LFO rate for H3000-style undulation. Rate sets the LFO speed, Depth sets how far the delay time is swept for pitch-wobble and chorus, and Swell controls the tremolo depth driven from a slow input follower. Mix blends the moving, detuned voice against the dry signal for shimmering, breathing modulation.

Param	Range	Default	Unit
Rate	0.05 – 5	0.5	Hz
Depth	0 – 1	0.5	—
Swell	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Delay

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FreqEcho 1 input

Frequency-shifted echo: a delay line whose feedback is single-sideband frequency-shifted via a polyphase Hilbert pair, so every repeat slides up or down in Hz rather than in pitch. Time sets the delay (1 to 1000 ms), Shift sets the linear frequency offset (-500 to +500 Hz), Feedback (0 to 0.95) sets repeat count, and Mix blends dry/wet. Because the shift is inharmonic, the tail detunes and smears with each pass for metallic, ever-evolving echoes.

Param	Range	Default	Unit
Time	1 – 1000	250	ms
Shift	-500 – 500	80	Hz
Feedback	0 – 0.95	0.6	—
Mix	0 – 1	0.4	—

Multitap 1 input

Multi-tap delay: the input is written into a circular buffer and read back at several time-spaced taps that are summed, averaged, and fed back into the buffer. Time sets the base delay in ms, Taps chooses 1 to 6 read points, Feedback recirculates the averaged taps for repeats, Spread pulls the taps toward the longer end for uneven echo timing, and Mix crossfades dry input against the wet sum. Low Spread gives evenly stacked echoes while high Spread spaces them out, and higher Feedback builds longer rhythmic delay trails.

Param	Range	Default	Unit
Time	1 – 1000	250	ms
Taps	1 – 6	3	—
Feedback	0 – 0.9	0.4	—
Spread	0 – 1	0.5	—
Mix	0 – 1	0.4	—

Bouncer 2 inputs

A feedback delay whose delay time continually shrinks, so the echoes speed up toward a stutter like a bouncing ball; a gate on in1 re-arms the delay back to its start time. Time sets the initial delay (20..600 ms), Accel sets how fast the time shrinks, Feedback sets the echo regeneration (up to 0.95), and Mix blends wet against dry. Higher Accel collapses the repeats into a rapid buzz, while a gate retrigger resets the bounce.

Param	Range	Default	Unit
Time	20 – 600	250	ms
Accel	0 – 1	0.5	—
Feedback	0 – 0.95	0.7	—
Mix	0 – 1	0.5	—

Beatmasher 2 inputs

Beat-repeat: while the gate on in1 is held it records a slice of audio on the first pass then loops that captured slice; releasing the gate passes the dry signal through. Length sets the size of the captured/looped slice (20..1000 ms) and Mix blends the looped slice against the dry input. Hold the gate to freeze and stutter a fragment, release to drop back to live audio.

Param	Range	Default	Unit
Length	20 – 1000	250	ms
Mix	0 – 1	1	—

BeatSlicer 2 inputs

Captures a buffer while the gate on in1 is held, divides it into 8 slices, and replays them in one of several reorder patterns; releasing the gate passes dry audio through. Length sets the buffer/loop size (100..2000 ms), Pattern selects the slice reordering (0..4, e.g. straight, swapped pairs, doubled, reversed, interleaved), and Mix blends the sliced output against dry. Hold the gate to scramble a phrase into a rhythmic rearrangement.

Param	Range	Default	Unit
Length	100 – 2000	500	ms
Pattern	0 – 4	1	—
Mix	0 – 1	1	—

ReverseGrain 2 inputs

Captures a buffer while the gate on in1 is held, then plays it backward at a pitch-shifted read rate for a reversed, stuttered texture; releasing the gate passes dry audio through. Length sets the captured grain size (50..1500 ms), Pitch transposes the reverse playback rate (+/-12 semitones), and Mix blends the reversed grain against dry. Hold the gate to suck a phrase into a backward swell.

Param	Range	Default	Unit
Length	50 – 1500	400	ms
Pitch	-12 – 12	0	st
Mix	0 – 1	1	—

PeakFlanger 1 input

A comb flanger whose delay time is swept by a peak envelope follower on the input, so transients drive the flange motion (delay ranges ~0.5..9.5 ms). Soften sets the envelope release time, Feedback sets the comb regeneration (+/-0.95, negative inverts the comb), Amount scales how far each peak sweeps the delay, and Mix blends wet against dry. The Traktor Flanger Pulse character: the jet sweep follows the music's own dynamics rather than a fixed LFO.

Param	Range	Default	Unit
Soften	0 – 1	0.5	—
Feedback	-0.95 – 0.95	0.5	—
Amount	0 – 1	0.7	—
Mix	0 – 1	0.5	—

SpaceEcho 3 inputs

Roland Space Echo-style tape delay: a wow-modulated tape head with saturated, tone-shaped feedback repeats.

Param	Range	Default	Unit
Time	50 – 700	300	ms
Feedback	0 – 0.95	0.4	—
Wow	0 – 1	0.2	—
Tone	1000 – 12000	5000	Hz
Mix	0 – 1	0.4	—

Echorec 3 inputs

Binson Echorec-style magnetic drum delay: several fixed playback heads give rhythmic multi-tap echoes with warm feedback.

Param	Range	Default	Unit
Time	40 – 350	150	ms
Heads	1 · 1-2 · 1-2-3 · All	—	—
Feedback	0 – 0.9	0.4	—
Mix	0 – 1	0.4	—

MemoryMan 3 inputs

EHX Memory Man-style analog BBD delay: dark, gently chorused repeats with companded feedback.

Param	Range	Default	Unit
Time	30 – 550	250	ms
Feedback	0 – 0.9	0.35	—
Chorus	0 – 1	0.3	—
Mix	0 – 1	0.4	—

SDD3000 3 inputs

Korg SDD-3000-style delay: a clean, bright digital delay with a characterful preamp on the input.

Param	Range	Default	Unit
Time	20 – 900	350	ms
Feedback	0 – 0.9	0.35	—
Tone	2000 – 16000	9000	Hz
Mix	0 – 1	0.4	—

TimeCube 3 inputs

Cooper Time Cube-style delay: a short, dark acoustic-tube delay (sound piped through a garden hose between transducers) for tight doubling and slapback. Time sets the short delay, Feedback the repeats and Mix the blend. Naturally band-limited and woody - characterful on vocals and snare, not a clean digital echo.

Param	Range	Default	Unit
Time	5 – 80	30	ms
Feedback	0 – 0.6	0.2	—
Mix	0 – 1	0.4	—

Echoplex 3 inputs

Maestro Echoplex EP-34-style tape echo: warm, wow-modulated repeats with tape saturation and HF loss.

Param	Range	Default	Unit
Time	80 – 650	300	ms
Feedback	0 – 0.9	0.4	—
Wow	0 – 1	0.3	—
Mix	0 – 1	0.4	—

Delay 3 inputs

Feedback delay line (1-2000 ms) with a damping low-pass in the feedback path that progressively darkens each repeat. Time is modulated by in 3 (scaled by Mod) for chorus and flange-style movement, Feedback sets regeneration up to 0.99, and Mix sets dry/wet. Damp at 0 gives bright digital echoes; higher values emulate tape/BBD high-frequency loss.

Param	Range	Default	Unit
Time	1 – 2000	250	ms
Feedback	0 – 0.99	0.35	—
Mod	0 – 500	0	ms
Damp	0 – 1	0	—
Mix	0 – 1	1	—

Comb 1 input

Tuned comb resonator: a short feedback delay set by Freq (20-2000 Hz) with a damping low-pass, which doubles as a Karplus-Strong plucked string when excited by a transient. Feedback sets resonance and sustain while Damp darkens the decay. High feedback rings at the tuned pitch; pair it with noise or an impulse for string and membrane tones.

Param	Range	Default	Unit
Freq	20 – 2000	220	Hz
Feedback	0 – 0.99	0.9	—
Damp	0 – 1	0.3	—
Mix	0 – 1	1	—

Chorus 1 input

Single-voice chorus: an LFO (Rate, Depth) modulates a short base Delay (1-30 ms) and the detuned copy is blended back via Mix to thicken and widen the source. Short delays with light depth give subtle doubling; longer and deeper settings approach vibrato and flange. See Ensemble for a three-voice string-machine variant.

Param	Range	Default	Unit
Rate	0.05 – 8	0.8	Hz
Depth	0 – 1	0.5	—
Delay	1 – 30	12	ms
Mix	0 – 1	0.5	—

Flanger 1 input

Swept-comb flanger: an LFO sweeps a very short delay (0.5-10 ms) mixed with the dry signal to create moving comb notches, with bipolar Feedback resonating the peaks (negative values invert for a hollow tone). Depth sets the sweep width and Mix the blend. Short delays with high feedback give the classic metallic jet-whoosh.

Param	Range	Default	Unit
Rate	0.05 – 8	0.3	Hz
Depth	0 – 1	0.7	—
Feedback	-0.95 – 0.95	0.3	—
Delay	0.5 – 10	2	ms
Mix	0 – 1	0.5	—

Vibrato 1 input

Pitch vibrato: an LFO modulates a short delay's read position to bend pitch, output fully wet (no dry blend) for true vibrato rather than chorus. Rate sets the wobble speed and Depth its pitch range. Gentle settings add expression; deep settings give a seasick warble.

Param	Range	Default	Unit
Rate	0.1 – 12	5	Hz
Depth	0 – 1	0.3	—
Delay	1 – 20	5	ms

Stutter 1 input

Auto-rhythmic buffer repeat: at each Rate cycle it snapshots a Grab-length window and loops it for the cycle, creating stutters and beat-repeat artefacts, with Mix blending against the live signal. Short grabs at high rates give granular buzz; longer grabs repeat recognizable slices. See Beatmasher and BeatSlicer for tempo-locked glitching.

Param	Range	Default	Unit
Rate	0.5 – 16	4	Hz
Grab	10 – 250	60	ms
Mix	0 – 1	1	—

TapeDelay 1 input

Vintage tape echo: a ~1.5 Hz wow modulates the delay time while a Tone low-pass and tanh tape Saturation in the feedback path darken and thicken each repeat. Feedback sets regeneration and Mix the blend. Unlike the clean Delay, repeats degrade and warble musically - raise Wow and Sat for worn-tape character.

Param	Range	Default	Unit
Time	1 – 2000	300	ms
Feedback	0 – 0.99	0.45	—
Wow	0 – 1	0.2	—
Tone	500 – 16000	5000	Hz
Sat	0 – 1	0.3	—
Mix	0 – 1	0.4	—

Ensemble 1 input

Three-voice string-machine chorus: one LFO modulates three delay taps at evenly spread phases, summing into a lush, wide thickening richer than the single-voice Chorus. Rate and Depth set the swirl and Mix the blend. Classic for pads, strings and synth-brass.

Param	Range	Default	Unit
Rate	0.05 – 4	0.6	Hz
Depth	0 – 1	0.6	—
Mix	0 – 1	0.5	—

Rotary 1 input

Leslie rotary speaker: combines a doppler pitch wobble (modulated short delay) with a phase-offset amplitude tremolo so the tone swirls as the virtual horn spins. Speed sets the rotation rate and Depth its intensity. Iconic on organ, electric piano and guitar - automate Speed for the slow-to-fast chorale ramp.

Param	Range	Default	Unit
Speed	0.5 – 8	6	Hz
Depth	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Reverse 1 input

Reverse delay: snapshots a Window-length slice of input and plays it backward on a loop, a glitchy pre-echo/reverse texture, blended via Mix. Short windows give fluttering grains; longer windows reverse recognizable phrases. See ReverseGrain and ReverseVerb for related effects.

Param	Range	Default	Unit
Window	20 – 500	120	ms
Mix	0 – 1	1	—

BBDDelay 1 input

Bucket-brigade style echo whose delayed tap is passed through a one-pole low-pass and a tanh saturation in the feedback path, so each repeat grows darker and grittier. Time sets the delay length (1-800 ms), Feedback sets regeneration up to 0.9, Tone sets the high-frequency rolloff of the repeats, and Mix blends the filtered echo with the dry signal. Lower Tone yields the murky, decaying character of analog BBD delays.

Param	Range	Default	Unit
Time	1 – 800	250	ms
Feedback	0 – 0.9	0.4	—
Tone	0 – 1	0.5	—
Mix	0 – 1	0.5	—

ReverseDelay 1 input

Captures the input into a delay buffer and reads it back newest-to-oldest within a repeating window, so each chunk plays in reverse. Time sets the window length (50-1000 ms) that defines how much audio is flipped per cycle, and Mix blends the reversed playback against the dry signal. Useful for backward-tape swells and disorienting rhythmic stutters.

Param	Range	Default	Unit
Time	50 – 1000	400	ms
Mix	0 – 1	0.5	—

DuckDelay 1 input

Feedback delay line whose wet output is attenuated by an envelope follower tracking the dry signal, so repeats drop out while you play and swell up in the gaps. Time sets the delay length (1-1000 ms), Feedback sets regeneration up to 0.9, Duck sets how hard the dry level suppresses the echoes, and Mix blends wet and dry. Keeps echoes from cluttering busy passages while letting them bloom in the silences.

Param	Range	Default	Unit
Time	1 – 1000	300	ms
Feedback	0 – 0.9	0.4	—
Duck	0 – 1	0.7	—
Mix	0 – 1	0.5	—

GrainDelay 1 input

Granular feedback delay that reads overlapping windowed grains from the delay line, pitch-shifting them and scattering their read positions. Time sets the base tap (10-1000 ms), Pitch transposes the grains +/-12 semitones, and Scatter randomizes each grain's offset for a smeared, diffuse cloud. Mix sets dry/wet; the shifted grains are fed back at reduced gain, so repeats drift in pitch and position.

Param	Range	Default	Unit
Time	10 – 1000	300	ms
Pitch	-12 – 12	0	st
Scatter	0 – 1	0.4	—
Mix	0 – 1	0.5	—

DiffuseDelay 1 input

Feedback delay whose tap is smeared through a 4-stage all-pass diffusion chain, pushing echoes toward reverb. Time sets the delay (1-1000 ms), Feedback sets regeneration up to 0.9, and Diffuse sets the all-pass coefficient that controls how much each repeat is blurred. Mix sets dry/wet; low Diffuse gives clean echoes, high Diffuse dissolves them into a diffuse wash.

Param	Range	Default	Unit
Time	1 – 1000	300	ms
Feedback	0 – 0.9	0.4	—
Diffuse	0 – 0.9	0.6	—
Mix	0 – 1	0.5	—

Looper 1 input

Circulating-buffer looper that re-reads a fixed-length loop and either overdubs new input onto it or recirculates it unchanged. Length sets the loop time (100-2000 ms) and Record switches between overdub (input mixed with the looped buffer at half gain) and pure recirculation when off. Mix sets dry/wet; leave Record on to layer takes, turn it off to freeze and play back the captured loop.

Param	Range	Default	Unit
Length	100 – 2000	1000	ms
Record	0 – 1	1	—
Mix	0 – 1	0.5	—

Shuffler 1 input

Beat-shuffler that records the input and, at each segment boundary, may jump its playback to one of a few earlier segment offsets. Segment sets the slice length (50-500 ms) and Chance sets the probability that a new boundary triggers a jump to a randomly chosen prior offset. Mix sets dry/wet; low Chance plays mostly straight, high Chance stutters and rearranges the beat.

Param	Range	Default	Unit
Segment	50 – 500	200	ms
Chance	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Scratch 1 input

Turntable-scratch effect where a raised-cosine LFO sweeps the delay read head back and forth through recent input. Rate sets the sweep speed (0.1-8 Hz) and Depth sets how far back the head travels (1-200 ms). Mix sets dry/wet; faster Rate and deeper Depth give more pronounced vinyl-scratch wobble.

Param	Range	Default	Unit
Rate	0.1 – 8	2	Hz
Depth	1 – 200	80	ms
Mix	0 – 1	0.5	—

PingPong 1 input

Ping-pong delay for a mono source: a single line is read at time T or $2T$ so the two outputs hear staggered taps that appear to bounce between speakers, with regeneration fed back through the longer tap. Time sets the base delay, Feedback the number of repeats up to 0.9, and Mix dry/wet. Out picks L or R, where R hears T and L hears $2T$ to create the alternating bounce.

Param	Range	Default	Unit
Time	1 – 1000	300	ms
Feedback	0 – 0.9	0.45	—
Out	0 – 1	0	—
Mix	0 – 1	0.5	—

Glitch 1 input

Probabilistic buffer glitcher that records the input and, at each slice boundary, randomly jumps to a captured slice and may play it forward or reversed. Slice sets the segment length in milliseconds and Chance sets the probability that a given slice is repeated or reversed rather than passed through cleanly. Mix blends the glitched playback against the dry signal for stutter, repeat and reverse artifacts.

Param	Range	Default	Unit
Slice	50 – 500	150	ms
Chance	0 – 1	0.4	—
Mix	0 – 1	1	—

Reverb

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Convolution 1 input

Convolution reverb: input is buffered into blocks and convolved with an impulse response via partitioned FFT, then streamed back with block latency. The IR comes from the node config (an audio file path loads that real IR, otherwise a decay time in seconds generates a decaying-noise tail); PreDelay offsets the wet by 0 to 200 ms, Tone is a one-pole low-pass on the wet, and Mix blends dry/wet. Use it to place sounds in real spaces or to apply any captured IR.

Param	Range	Default	Unit
Mix	0 – 1	0.3	—
PreDelay	0 – 200	0	ms
Tone	0 – 1	1	—

Plate 1 input

Plate reverb: the input runs through four series allpass diffusers and then into four parallel damped feedback combs that are summed for the wet tail. Decay sets the comb feedback (0 to 0.95), Damp is a one-pole low-pass inside the feedback that darkens later reflections, Size scales the delay-line lengths, and Mix blends dry/wet. Dense, smooth metallic decay suited to plates and tight ambiences.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Decay	0 – 0.95	0.7	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.3	—

Shimmer 1 input

Shimmer reverb: a delay line whose feedback path is pitch-shifted by a two-window granular shifter so repeats climb in pitch each pass. Shift sets the transposition in semitones (-12 to +24), Feedback (0 to 0.95) controls how long the tail sustains, Size sets the delay time in ms, Damp low-passes the feedback, and Mix blends dry/wet. With Shift at +12 it builds the classic ethereal rising-octave wash.

Param	Range	Default	Unit
Size	50 – 800	300	ms
Shift	-12 – 24	12	st
Feedback	0 – 0.95	0.7	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.4	—

Iceverb 1 input

Four lowpass-damped comb filters form the reverb body, then a self-oscillating resonant state-variable filter is applied to the tail for a cold, ringing sheen. Size sets the comb feedback/decay (up to 0.95), Color sets the resonant filter frequency (~100 Hz..14 kHz), Icing sets its resonance toward self-oscillation, and Mix blends wet against dry. High Icing makes the tail sing into a metallic, icy ring.

Param	Range	Default	Unit
Size	0 – 0.95	0.7	—
Color	0 – 1	0.5	—
Icing	0.5 – 14	8	—
Mix	0 – 1	0.35	—

ModEchoVerb 1 input

Fused reverb and modulated delay: the input feeds a wow-modulated feedback delay and a short comb reverb tank at once, blending echo and space in one block.

Param	Range	Default	Unit
Size	0 – 0.95	0.6	—
Time	20 – 1000	300	ms
Feedback	0 – 0.9	0.4	—
ModRate	0.05 – 5	0.4	Hz
Mix	0 – 1	0.4	—

EMT140 3 inputs

EMT 140-style plate reverb: instant dense diffusion with no early reflections, a bright sheen and a high end that decays faster than the low tail.

Param	Range	Default	Unit
Decay	0 – 1	0.6	—
Tone	0 – 1	0.4	—
PreDelay	0 – 200	0	ms
Mix	0 – 1	0.3	—

Chamber 3 inputs

Chamber reverb: higher modal density than a plate with a diffuse onset and warm, non-metallic tail - between a plate and a room.

Param	Range	Default	Unit
Decay	0 – 1	0.6	—
Tone	0 – 1	0.6	—
PreDelay	0 – 200	10	ms
Mix	0 – 1	0.3	—

BX20 3 inputs

AKG BX-20-style torsional-spring reverb: plate-like onset and chamber density with the spring chirp smeared smooth - long, dark, no boing.

Param	Range	Default	Unit
Decay	0 – 1	0.7	—
Tone	0 – 1	0.7	—
Mix	0 – 1	0.3	—

Lex224 3 inputs

Lexicon 224-style digital hall: a long, lush, slowly-modulated reverb tail (chorusing avoids metallic ringing) with bright, dense diffusion.

Param	Range	Default	Unit
Decay	0 – 1	0.7	—
Tone	0 – 1	0.5	—
PreDelay	0 – 200	20	ms
Mix	0 – 1	0.3	—

Lex480 3 inputs

Lexicon 480L-style digital reverb: a brighter, denser random-hall/plate with a modulated tail and fast build-up.

Param	Range	Default	Unit
Decay	0 – 1	0.65	—
Tone	0 – 1	0.6	—
PreDelay	0 – 200	10	ms
Mix	0 – 1	0.3	—

EMT250 3 inputs

EMT 250-style early digital reverb: bright, dense and grainy with the unmistakable first-generation digital character.

Param	Range	Default	Unit
Decay	0 – 1	0.55	—
Tone	0 – 1	0.55	—
PreDelay	0 – 150	8	ms
Mix	0 – 1	0.3	—

RMX16 3 inputs

AMS RMX16-style digital reverb (Nonlin program): a dense, gated tail that holds then cuts off instead of decaying naturally - the classic 80s gated sound.

Param	Range	Default	Unit
Density	0 – 1	0.6	—
Hold	20 – 400	150	ms
Tone	0 – 1	0.5	—
Mix	0 – 1	0.3	—

Reverb 1 input

Algorithmic room reverb (Freeverb-style comb/all-pass network) with Size scaling the decay time and Damp rolling off the highs in the tail. Width sets the stereo spread and Mix the dry/wet blend. Large Size with low Damp gives long bright halls; high Damp yields short dark rooms - see Massive, Plate, Spring and Shimmer for character-specific spaces.

Param	Range	Default	Unit
Size	0 – 1	0.5	—
Damp	0 – 1	0.5	—
Width	0 – 1	1	—

Param	Range	Default	Unit
Mix	0 – 1	0.3	—

Massive 1 input

Feedback-delay-network reverb (Valhalla Supermassive style): four modulated delay lines mixed through a unitary Hadamard matrix give huge, smooth, near-infinite tails. Size scales delay length up to multi-second, Decay sets feedback, Damp absorbs highs in the loop, and Mod choruses the lines for shimmer. Mode picks Lush / Huge / Cosmos voicings, from tight to cathedral to cosmic.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Decay	0 – 1	0.85	—
Damp	0 – 1	0.4	—
Mod	0 – 1	0.3	—
Mode	0 – 2	1	—
Mix	0 – 1	0.4	—

Spring 1 input

Spring-reverb approximation: a short (~30 ms) delay loop diffused through an all-pass cascade for the characteristic dispersive boing, with a Tone low-pass in the feedback. Decay sets the tail length and Mix the blend. The classic guitar-amp and dub reverb - splashy on transients, drippy at high Decay.

Param	Range	Default	Unit
Decay	0 – 0.95	0.6	—
Tone	500 – 8000	3000	Hz
Mix	0 – 1	0.3	—

Ascend 1 input

Pitch-rising shimmer reverb: a long delay whose feedback is pitch-shifted up an octave (granular two-tap) and damped, so the tail ascends and blooms into ethereal pads. Size sets the delay length, Shimmer the octave-up feedback amount and Damp the tail brightness. Lush on pads, vocals and ambient guitar - see Crystals for reverse+pitch and Gravity for huge inverted tails.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Shimmer	0 – 0.95	0.5	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.3	—

GatedVerb 1 input

Reverb whose wet tail is muted by a fast envelope-driven gate, so the reverb only passes while the dry input stays above a threshold. Size sets the room length and Damp the high-frequency decay, Threshold sets the level below which the tail is cut, and Mix blends the gated reverb against the dry signal. Reproduces the abrupt, chopped-off reverb of 80s drum production.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Threshold	-60 – 0	-30	dB
Mix	0 – 1	0.3	—

FreezeVerb 1 input

Reverb that can latch into an infinite-hold freeze mode, sustaining the current tail indefinitely instead of decaying. Size sets the room length and Damp the high-frequency decay, Freeze above its midpoint engages the infinite hold, and Mix blends the reverb against the dry signal. Use the freeze to build sustained drones and pads from a single moment of input.

Param	Range	Default	Unit
Size	0 – 1	0.7	—
Damp	0 – 1	0.3	—
Freeze	0 – 1	0	—
Mix	0 – 1	0.4	—

DuckVerb 1 input

Reverb whose wet signal is attenuated by an envelope follower tracking the dry input, so the tail dips while you play and swells back in the gaps. Size sets the room length and Damp the high-frequency decay, Duck sets how strongly the dry level suppresses the wet, and Mix blends the ducked reverb against the dry signal. Keeps the reverb from washing out a busy performance while preserving the tail between phrases.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Duck	0 – 1	0.7	—
Mix	0 – 1	0.3	—

ModVerb 1 input

Reverb whose room size is continuously modulated by a sine LFO, detuning the tail for a chorused, shimmering character. Size sets the base room length and Damp the high-frequency decay, Rate sets the LFO speed (0.05-5 Hz) and Depth how far it sweeps the size, and Mix blends the moving reverb against the dry signal. Adds motion and width to otherwise static reverb tails.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Rate	0.05 – 5	0.5	Hz
Depth	0 – 1	0.3	—
Mix	0 – 1	0.3	—

ReverseVerb 1 input

Reverb whose wet level swells up from silence after each detected onset, fading the tail in rather than out for a backward, blooming effect. Size sets the room length and Damp the high-frequency decay, Swell sets the fade-in time (1-500 ms) restarted on every new transient, and Mix blends the swelling reverb against the dry signal. Produces the rising, sucked-in quality of reversed reverb.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Swell	1 – 500	150	ms
Mix	0 – 1	0.4	—

Gravity 1 input

Oversized 4-line feedback delay network (FDN) reverb, each line damped by a low-pass and length-modulated by a shared LFO. Size stretches the base delay times up to 9x, Gravity raises feedback toward unity for near-infinite tails, Mod sets the LFO modulation

depth, and Freeze pins feedback to 1.0 while muting new input. Mix blends the Hadamard-mixed wet sum against the dry; pushed hard it yields huge, slowly-evolving, freezable spaces.

Param	Range	Default	Unit
Size	0 – 1	0.8	—
Gravity	0 – 1	0.9	—
Mod	0 – 1	0.4	—
Freeze	0 – 1	0	—
Mix	0 – 1	0.4	—

NonlinVerb 1 input

Nonlinear 4-line FDN reverb whose wet tail is gated by an envelope follower: a transient above the running level retriggers a hold, then the gain holds flat before cutting off. Size scales the short base delay times up to 5x, while Hold sets how long (10-500 ms) the tail sustains at full level before it decays away. Mix sets dry/wet; the held-then-cut behaviour gives gated, non-natural reverb bursts rather than a smooth decay.

Param	Range	Default	Unit
Size	0 – 1	0.5	—
Hold	10 – 500	150	ms
Mix	0 – 1	0.4	—

Crystals 1 input

Shimmer reverb: a 4-line FDN whose output is pitch-shifted an octave up and re-injected into the network, building rising crystalline overtones. Size scales the base delay times up to 6x, Shimmer sets how much octave-up signal feeds back in, and Damp sets the per-line low-pass that darkens repeats. Mix blends the wet sum against the dry; higher Shimmer ascends further, higher Damp tames the brightness.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Shimmer	0 – 1	0.5	—
Damp	0 – 1	0.4	—
Mix	0 – 1	0.4	—

GrainVerb 1 input

Granular reverb that windows the input with a moving Hann grain and injects it into a 4-line feedback delay network for diffusion. Size scales the four delay lengths up to 6x for longer tails, Grain sets the grain rate in milliseconds, and Damp applies a low-pass in each feedback line to darken successive reflections. Mix blends the diffused tail against the dry input for a smeared, granular ambience.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Grain	5 – 100	30	ms
Damp	0 – 1	0.4	—
Mix	0 – 1	0.3	—

Freezer 1 input

Audio freezer that continuously buffers the input and, when Freeze is engaged, loops the most recently captured window indefinitely. Size sets the length of the looped window in milliseconds, trading a short stuttering grain against a longer evolving pad, and Mix blends the frozen loop against the dry signal. With Freeze off it passes the input through while keeping the buffer primed for an instant capture.

Param	Range	Default	Unit
Freeze	0 – 1	0	—
Size	20 – 500	200	ms
Mix	0 – 1	1	—

MangledVerb 1 input

Reverb whose fully wet output is fed through a tanh saturation stage for gritty, broken ambience. Size sets the room size and tail length, Damp rolls off high frequencies in the decay, and Drive pushes the reverb output harder into the nonlinearity for increasing crush. Mix blends the distorted tail against the dry signal, ranging from a lightly dirtied space to a heavily mangled wash.

Param	Range	Default	Unit
Size	0 – 1	0.6	—
Damp	0 – 1	0.4	—
Drive	0 – 1	0.5	—
Mix	0 – 1	0.3	—

Dynamics

30 modules

Omnipressor 1 input

Extreme dynamics on one knob: Function sweeps from gating expansion through limiting compression to full dynamic reversal, where loud passages turn quiet and quiet turns loud.

Param	Range	Default	Unit
Function	-1 – 3	1	—
Threshold	-48 – 0	-18	dB
Attack	0.1 – 100	3	ms
Release	5 – 800	120	ms
Gain	-12 – 24	0	dB

VariMu 3 inputs

Fairchild/Manley-style tube vari-mu compressor: program-dependent ratio rises with level, soft knee, tube even-harmonic warmth.

Param	Range	Default	Unit
Threshold	-48 – 0	-18	dB
Ratio	1.5 – 8	3	—
Attack	1 – 100	20	ms
Release	50 – 2000	400	ms
Makeup	-6 – 24	0	dB

Opto 3 inputs

LA-2A-style optical compressor: soft fixed-ratio leveling with a two-stage (fast then slow) program-dependent release; smooth and gentle.

Param	Range	Default	Unit
Threshold	-48 – 0	-20	dB
Ratio	2 – 6	3	—
Attack	1 – 50	10	ms
Release	50 – 3000	600	ms
Makeup	-6 – 24	0	dB

FET 3 inputs

1176-style FET compressor: ultra-fast attack, fixed high ratios, asymmetric FET harmonic bite (max ratio = all-buttons slam).

Param	Range	Default	Unit
Threshold	-48 – 0	-16	dB
Ratio	4 – 20	8	—
Attack	0.02 – 5	0.5	ms
Release	20 – 1200	200	ms

Param	Range	Default	Unit
Makeup	-6 – 24	0	dB

SSL Bus 3 inputs

SSL bus-style VCA glue compressor: a dual-time-constant program-dependent auto-release (fast for transients, slow under sustained level) for pumping-free mix glue.

Param	Range	Default	Unit
Threshold	-40 – 0	-14	dB
Ratio	2 – 10	4	—
Attack	0.1 – 30	10	ms
Release	0.1 – 1.2	0.3	s
Makeup	-6 – 24	0	dB

Dbx160 3 inputs

dbx 160-style VCA compressor: true-RMS detection and an 'over-easy' soft knee make the gain change nearly inaudible; punchy and musical.

Param	Range	Default	Unit
Threshold	-40 – 0	-18	dB
Ratio	1 – 20	4	—
Knee	0 – 20	12	dB
Makeup	-6 – 24	0	dB

Distressor 3 inputs

Distressor-style aggressive compressor with switchable harmonic distortion (Dist2 = even/2nd, Dist3 = odd/3rd) and a British-mode all-buttons slam.

Param	Range	Default	Unit
Threshold	-40 – 0	-16	dB
Ratio	1 – 20	6	—
Attack	0.05 – 5	0.5	ms
Release	20 – 1200	150	ms
Dist	Clean · Dist2 · Dist3	—	—
British	Off · On	—	—

LA3A 3 inputs

Teletronix LA-3A-style optical compressor: the same opto cell as the LA-2A but a faster, brighter solid-state amp - punchier optical leveling.

Param	Range	Default	Unit
Threshold	-40 – 0	-18	dB
Ratio	2 – 8	4	—
Makeup	-6 – 24	0	dB
Mix	0 – 1	1	—

API2500 3 inputs

API 2500-style VCA bus compressor: a Thrust sidechain tilt (desensitises the detector to bass) with Old (feedback) or New (feedforward) detection.

Param	Range	Default	Unit
Threshold	-40 – 0	-14	dB
Ratio	1.5 – 10	4	—
Attack	0.1 – 30	10	ms
Release	50 – 2000	300	ms
Thrust	Off · Med · Loud		—
Mode	Old · New		—

StaLevel 3 inputs

Gates Sta-Level-style tube vari-mu limiter: a very slow, program-dependent auto-release and gentle ~3:1 leveling with even-harmonic tube warmth.

Param	Range	Default	Unit
Threshold	-40 – 0	-16	dB
Release	0 – 1	0.5	—
Makeup	-6 – 24	0	dB
Warmth	0 – 1	0.3	—

Neve33609 3 inputs

Neve 33609-style diode-bridge bus compressor: smooth VCA-style glue with a dual-time-constant release and transformer warmth.

Param	Range	Default	Unit
Threshold	-40 – 0	-16	dB
Ratio	1.5 – 6	3	—
Attack	0.1 – 30	8	ms
Release	50 – 2000	400	ms
Makeup	-6 – 24	0	dB

TubeTech 3 inputs

Tube-Tech CL 1B-style opto compressor: slow, smooth optical leveling with adjustable attack/release and tube warmth.

Param	Range	Default	Unit
Threshold	-40 – 0	-18	dB
Ratio	1.5 – 10	4	—
Attack	5 – 300	50	ms
Release	100 – 5000	1000	ms
Makeup	-6 – 24	0	dB

ShadowHills 3 inputs

Shadow Hills Mastering-style compressor: an optical stage into a discrete VCA stage, with a transformer-flavour select (nickel/iron/steel).

Param	Range	Default	Unit
Threshold	-40 – 0	-16	dB
Opto Ratio	1.5 – 4	2	—
VCA Ratio	2 – 8	3	—
Release	100 – 2000	500	ms
Xfmr	Nickel · Iron · Steel		—
Makeup	-6 – 24	0	dB

Summit 3 inputs

Summit TLA-100-style tube/opto leveling amplifier: smooth optical compression with tube makeup warmth.

Param	Range	Default	Unit
Threshold	-40 – 0	-18	dB
Ratio	2 – 10	4	—
Attack	1 – 100	20	ms
Release	100 – 3000	600	ms
Makeup	-6 – 24	0	dB

SCComp 2 inputs

Sidechain compressor: compresses the signal (in 0) by the level of a key input (in 1) - cable the host sidechain (SC L/R) to the key for classic ducking/pumping.

Param	Range	Default	Unit
Threshold	-48 – 0	-18	dB
Ratio	1 – 20	4	—
Attack	0.1 – 100	5	ms
Release	1 – 500	100	ms
Makeup	-6 – 24	0	dB

Envelope 1 input

Envelope follower that rectifies the input and smooths it with independent Attack and Release times into a control signal. Gain scales the output for driving a filter cutoff, VCA or any mod input (auto-wah, ducking, dynamics). Fast attack tracks transients; a longer release gives a smoother contour.

Param	Range	Default	Unit
Attack	0.1 – 200	5	ms
Release	1 – 1000	80	ms
Gain	0 – 8	1	—

Compressor 1 input

Peak-following compressor: tracks the input envelope with Attack/Release ballistics and, above Threshold, reduces gain by the set Ratio, with Makeup to restore level. Fast attack tames transients while slow release glues sustained material. A clean general-purpose dynamics stage - see the analog pack (FET, Opto, VariMu) for coloured compression.

Param	Range	Default	Unit
Threshold	-60 – 0	-18	dB
Ratio	1 – 20	4	—
Attack	0.1 – 100	5	ms

Param	Range	Default	Unit
Release	1 – 500	80	ms
Makeup	-12 – 24	0	dB

Gate 1 input

Noise gate: when the detected level falls below Threshold it attenuates the signal toward Range (the floor), opening and closing with Attack/Release ballistics. Range at -80 dB fully mutes between events; smaller values leave a residual for gentler ducking. Use it to silence hiss and bleed between phrases or for rhythmic gating.

Param	Range	Default	Unit
Threshold	-80 – 0	-40	dB
Attack	0.1 – 50	1	ms
Release	1 – 500	100	ms
Range	-80 – 0	-80	dB

Limiter 1 input

Brick-wall peak limiter: instantly clamps gain so peaks never exceed Threshold, then releases smoothly back to unity over Release, with Output trimming the post-limit level. Set Threshold as the ceiling and drive the input for transparent loudness. See Maximizer for loudness-optimized limiting.

Param	Range	Default	Unit
Threshold	-24 – 0	-3	dB
Release	1 – 500	50	ms
Output	-12 – 12	0	dB

Transient 1 input

Transient shaper: a fast vs slow envelope follower separates the attack (fast > slow) from the sustain (fast < slow), letting Attack and Sustain each be boosted or cut independently, with Output trimming the result. Add punch to drums, or tighten/lengthen sustain - all independent of input level, unlike a compressor.

Param	Range	Default	Unit
Attack	-1 – 1	0.3	—
Sustain	-1 – 1	0	—
Output	-12 – 12	0	dB

Expander 1 input

Downward expander: attenuates signal that falls below Threshold by the set Ratio (the inverse of a compressor), tightening quiet passages and reducing bleed and hiss between notes. Attack/Release set the ballistics. Gentler than a gate - it fades low-level material rather than hard-muting it.

Param	Range	Default	Unit
Threshold	-60 – 0	-40	dB
Ratio	1 – 10	2	—
Attack	0.1 – 100	5	ms
Release	1 – 500	100	ms

MultiComp 1 input

Three-band compressor: crossovers split the signal into low, mid and high bands (Low/Mid and Mid/High), each compressed independently against a shared Threshold and Ratio with fast-attack/slow-release ballistics, then summed with Makeup. Tame a busy mix's bands separately - control boomy lows without dulling the highs. See the SSL/dbx analog models for coloured bus compression.

Param	Range	Default	Unit
Low/Mid	100 – 1000	300	Hz
Mid/High	1000 – 12000	3000	Hz
Threshold	-48 – 0	-18	dB
Ratio	1 – 20	4	—
Makeup	-12 – 24	0	dB

DeEsser 1 input

Tames sibilance by splitting off a high band with a one-pole low-pass, tracking its level with a fast-attack envelope, and applying gain reduction to that band only when it crosses the threshold before recombining it with the lows. Freq sets the sibilance band, Threshold sets where reduction begins, Amount scales how hard the high band is pulled down, and Mix sets dry/wet. Because only the highs are compressed, it removes harsh esses without dulling the rest of the signal.

Param	Range	Default	Unit
Freq	2000 – 12000	6000	Hz
Threshold	-60 – 0	-24	dB
Amount	0 – 1	0.7	—
Mix	0 – 1	1	—

Ducker 2 inputs

Sidechain ducker that follows the level of the key input (in 2) with a release-smoothed envelope and pulls down the main signal whenever that key exceeds the threshold. Threshold sets the key level that triggers ducking, Amount sets how far the main signal drops, and Release sets how quickly it recovers once the key falls back. Use it for music-under-voice ducking or rhythmic pumping driven by a separate trigger.

Param	Range	Default	Unit
Threshold	-60 – 0	-30	dB
Amount	0 – 1	0.8	—
Release	10 – 1000	200	ms

Maximizer 1 input

Loudness maximizer that applies a gain boost, tracks the peak envelope, and divides back any sample exceeding the ceiling before hard-clamping to it, acting as a brick-wall limiter. Boost sets how hard the input is driven up and Ceiling sets the absolute output peak in dB. Push Boost while keeping Ceiling near 0 dB to raise apparent loudness without overshooting the output.

Param	Range	Default	Unit
Boost	0 – 24	6	dB
Ceiling	-12 – 0	-0.3	dB

UpComp 1 input

Upward compressor that tracks the input envelope and, when it sits below the threshold, applies makeup gain proportional to how far below it is and to the ratio, lifting quiet passages toward the threshold. Threshold sets the level below which boosting begins,

Ratio sets how aggressively the deficit is filled, Makeup adds a fixed output gain, and Mix sets dry/wet. Use it to raise low-level detail and tighten dynamic range from the bottom up rather than taming peaks.

Param	Range	Default	Unit
Threshold	-60 – 0	-30	dB
Ratio	1 – 8	2	—
Makeup	0 – 24	0	dB
Mix	0 – 1	1	—

DynEQ 1 input

Dynamic EQ band: an SVF band-pass at Freq/Q tracks an envelope follower, and when that band's energy crosses Threshold the band is subtracted from the signal in proportion to Amount. Freq and Q place and narrow the band, Threshold sets where reduction begins, and Mix blends the result against dry. It only acts when the band gets loud, so it tames resonances and harshness without static EQ cuts.

Param	Range	Default	Unit
Freq	100 – 12000	2000	Hz
Q	0.5 – 8	2	—
Threshold	-60 – 0	-24	dB
Amount	0 – 1	0.7	—
Mix	0 – 1	1	—

MBDynamics 1 input

Multiband dynamics processor that splits the input into low, mid and high bands (crossovers near 250 Hz and 2.5 kHz) and applies independent compression and gating to each. Threshold sets the level above which a band is compressed at Ratio, while Gate sets the level below which a band is attenuated downward. Mix blends the processed sum against the dry signal, useful for tightening or de-noising specific frequency regions at once.

Param	Range	Default	Unit
Threshold	-60 – 0	-24	dB
Ratio	1 – 20	3	—
Gate	-80 – -20	-60	dB
Mix	0 – 1	1	—

Pump 1 input

Tempo-synced sidechain-style volume pump that ducks the input each cycle and recovers using a sample-clock phase for coherent timing. Rate sets the duck frequency in Hz, Depth sets how far the level drops, and Curve shapes the recovery ramp from snappy to gradual. Mix blends the pumped signal against the dry input for the classic EDM ducking effect without an external sidechain.

Param	Range	Default	Unit
Rate	0.1 – 8	2	Hz
Depth	0 – 1	0.8	—
Curve	0.2 – 4	1.5	—
Mix	0 – 1	1	—

Squash 1 input

Aggressive parallel smash: a fast peak-following envelope drives heavy gain reduction above a fixed threshold, and the gain-reduced signal is pushed through a tanh saturator for crushed, distorted dynamics. Amount steepens the gain-reduction curve (harder squash), Tone scales the saturated output level, and Mix blends the smashed signal in parallel with the dry input. Useful for adding punch and grit to drums and bus material.

Param	Range	Default	Unit
Amount	0 – 1	0.7	—
Tone	0 – 1	0.5	—
Mix	0 – 1	0.5	—

Pitch

13 modules

FreqShift 1 input

Bode-style single-sideband frequency shifter: splits the input at in0 into in-phase and 90-degree branches via a polyphase IIR Hilbert allpass pair, then heterodynes against an internal carrier to shift every partial by a fixed Hz amount. Shift sets the offset (-1000 to +1000 Hz) and Mix blends the shifted signal against the dry input. Because the shift is in Hz rather than ratio, harmonic relationships break, giving inharmonic, metallic, clangorous tones.

Param	Range	Default	Unit
Shift	-1000 – 1000	100	Hz
Mix	0 – 1	1	—

Spinback 2 inputs

Turntable brake: a gate on in1 starts reading the captured buffer at steadily falling speed, so playback drops in pitch and slows to a dead stop like braking a record. Time sets how long the brake takes to reach a halt (50..3000 ms) and Mix blends the braked signal against the dry input. Releasing the gate stops braking and returns to live audio.

Param	Range	Default	Unit
Time	50 – 3000	600	ms
Mix	0 – 1	1	—

H910 1 input

Harmonizer: a pitch shifter feeding a feedback delay, so each repeat is pitched again into a rising or falling cascade (the classic H910 fused pitch+delay).

Param	Range	Default	Unit
Pitch	-12 – 12	7	st
Delay	1 – 1000	200	ms
Feedback	0 – 0.9	0.4	—
Mix	0 – 1	0.5	—

Octavox 1 input

A cloud of four pitch-shifted voices at diatonic intervals, each on its own delay tap, summed into a shimmering harmonized pitch-delay (Octavox/Quadravox class).

Param	Range	Default	Unit
Spread	10 – 400	120	ms
Key	0 – 2	0	—
Feedback	0 – 0.85	0.3	—
Level	0 – 1	0.6	—
Mix	0 – 1	0.5	—

H3000 3 inputs

Eventide H3000-style micro-pitch doubler: two slightly detuned, short-delayed voices thicken the signal into a wide, lush double-track.

Param	Range	Default	Unit
Detune	0 – 30	12	ct
Delay	1 – 50	18	ms
Width	0 – 1	0.7	—
Mix	0 – 1	0.5	—

PitchShift 1 input

Granular pitch shifter using two overlapping, cross-faded grains (Window 20-120 ms) to transpose +/-24 semitones without changing length. Feedback re-injects the shifted output for cascading and shimmer effects, and Mix sets dry/wet. Shorter windows tighten transients but add warble; longer windows smooth the tone at the cost of latency.

Param	Range	Default	Unit
Pitch	-24 – 24	7	st
Mix	0 – 1	0.5	—
Feedback	0 – 0.9	0	—
Window	20 – 120	60	ms

Octaver 1 input

Sub-octave generator: a flip-flop toggled at each rising zero crossing produces a half-rate square tracked by the input envelope, low-passed by Tone for warmth and mixed under the Dry signal at Sub. Monophonic, analog-pedal style - best on clean single notes and bass. It glitches on chords, as expected of the technique.

Param	Range	Default	Unit
Sub	0 – 1	0.7	—
Dry	0 – 1	1	—
Tone	200 – 8000	2000	Hz

Unison 1 input

Unison thickener: turns a mono input into up to 16 pitch-detuned voices (the supersaw sound) using overlapping ramped-delay pitch shifters at symmetric cent offsets, so the centre stays at pitch and the outer voices spread apart. Detune sets the spread, Blend the outer-voice level and Mix dry/wet. Wide and lush on synths, pads and backing vocals.

Param	Range	Default	Unit
Voices	1 – 16	4	—
Detune	0 – 50	18	ct
Blend	0 – 1	0.7	—
Mix	0 – 1	1	—

MicroShift 1 input

Micro-pitch doubler that generates two delay-line voices detuned slightly up and slightly down, then sums them with the dry signal to widen the source. Detune sets the symmetric pitch offset (0-50 cents) and Delay sets the base tap time (0-40 ms) before the shifted voices. Mix sets dry/wet; small Detune thickens, larger Detune broadens into a chorus-like spread.

Param	Range	Default	Unit
Detune	0 – 50	12	cents
Delay	0 – 40	12	ms
Mix	0 – 1	0.5	—

Harmonizer 1 input

Single-voice harmonizer that pitch-shifts the input by a chosen musical interval using overlapping windowed delay-line reads, then adds it back to the dry signal. Interval sets the transposition (+/-24 semitones), Detune adds up to 30 cents of offset for a less rigid blend, and Level sets the harmony voice gain. Mix scales the added voice into the output; useful for parallel harmonies and octave doubling.

Param	Range	Default	Unit
Interval	-24 – 24	7	st
Detune	0 – 30	0	cents
Level	0 – 1	0.7	—
Mix	0 – 1	1	—

Pitch2Midi 1 input

Monophonic pitch tracker that measures the period between rising zero-crossings, converts it to a MIDI note number, and emits it as a control value. Glide smooths the tracked pitch over time for portamento, while Out switches the output between the normalized pitch CV and a simple gate that goes high whenever the input is active. Intended to drive other modules from a played signal rather than to process audio.

Param	Range	Default	Unit
Glide	0 – 0.99	0.3	—
Out	0 – 1	0	—

PitchCorrect 1 input

Rough auto-tune that detects the input period via zero-crossings, finds the nearest semitone, and applies a crossfaded two-tap windowed pitch shift to pull the note onto pitch. Amount scales how far the signal is moved toward the target semitone, from no correction to full snapping, and Mix blends the corrected voice against the dry input. Best on clean monophonic sources; artifacts increase with stronger correction.

Param	Range	Default	Unit
Amount	0 – 1	0.8	—
Mix	0 – 1	1	—

TimeStretch 1 input

Granular time-smear that writes the input into a delay line and reads it back with two overlapping Hann-windowed grains advancing slower than real time. Stretch sets how slowly the read position moves (lower values smear the sound out further) and Grain sets the grain length in milliseconds, trading smoothness against fluttering artifacts. Mix blends the stretched texture against the dry signal for frozen, drifting timbres.

Param	Range	Default	Unit
Stretch	0.25 – 1	0.5	—
Grain	20 – 200	80	ms
Mix	0 – 1	1	—

Utility

30 modules

Expr 3 inputs

User-written per-sample expression compiled by an RT-safe VM (P0..P5 params plus two signal inputs) - does what the fixed blocks can't.

Param	Range	Default	Unit
P0	0 – 1	0.5	—
P1	0 – 1	0.5	—
P2	0 – 1	0.5	—
P3	0 – 1	0.5	—
P4	0 – 1	0.5	—
P5	0 – 1	0.5	—

Gain 1 input

Gain trim (dB) plus a DC bias offset.

Param	Range	Default	Unit
Gain	-60 – 24	0	—
Bias	-1 – 1	0	—

Mixer 3 inputs

Sums three scaled inputs plus a DC offset.

Param	Range	Default	Unit
In 1	-1.5 – 1.5	1	—
In 2	-1.5 – 1.5	1	—
Mod Amt	-1.5 – 1.5	0	—
Offset	-1 – 1	0	—

Subtract 3 inputs

Difference node: in0 minus in1 minus in2, plus a DC Offset. Feed two signals to get their difference (error/side-chain CV, mid-from-sum, delta detectors) or leave in2 unpatched for a simple two-input subtract. Unconnected inputs contribute 0.

Param	Range	Default	Unit
Offset	-2 – 2	0	—

Multiply 2 inputs

Two-quadrant multiplier / VCA: in0 times in1 times Scale. With an audio-rate in1 it is ring modulation; with a control-rate in1 it is a clean amplitude VCA. Both inputs must be patched - an unconnected input is 0, which mutes the output (the multiplicative identity is not assumed).

Param	Range	Default	Unit
Scale	-4 – 4	1	—

Divide 2 inputs

Safe divider: in0 divided by (in1 + Denom), with the result limited to +/-Limit so a near-zero denominator can never blow up. Denom both seeds the divisor when in1 is unpatched and biases it when it is, and the magnitude of the divisor is floored at a tiny epsilon. Useful for normalising, ratio CV, and reciprocal-style shaping.

Param	Range	Default	Unit
Denom	-4 – 4	1	—
Limit	1 – 64	16	—

Negate 1 input

Sign inverter: outputs -in0 scaled by Scale (default 1) plus an Offset. Flip the phase of an audio signal, invert a control voltage, or build a difference by summing a Negate against another source. Scale at -1 passes the signal through unchanged.

Param	Range	Default	Unit
Scale	-2 – 2	1	—
Offset	-1 – 1	0	—

Reciprocal 1 input

One-over-x: $1 / (\text{in0} + \text{Bias})$, limited to +/-Limit and floored away from zero so it stays finite. Turns a rising CV into a falling one, converts frequency to period (and back), and builds hyperbolic shapes. Bias keeps the denominator off zero when in0 passes through the origin.

Param	Range	Default	Unit
Bias	-2 – 2	0	—
Limit	1 – 64	16	—

Min 2 inputs

Outputs the smaller of in0 and in1 (after adding the In B offset to in1). Pairs with Max to build hard clippers, envelope floors, and CV limiters. An unconnected in1 reads 0, so In B sets the constant floor in that case.

Param	Range	Default	Unit
In B	-2 – 2	0	—

Max 2 inputs

Outputs the larger of in0 and in1 (after adding the In B offset to in1). Max against 0 is a half-wave rectifier / positive-only clamp; against a constant it is an envelope ceiling. An unconnected in1 reads 0, so In B sets the constant ceiling in that case.

Param	Range	Default	Unit
In B	-2 – 2	0	—

Clamp 1 input

Hard range limiter: constrains in0 to the [Lo, Hi] window with no saturation curve (unlike Clip, which rounds the knee). Use it to bound CV, keep a feedback signal in range, or window a modulator. If Lo > Hi the bounds swap so the window is always valid.

Param	Range	Default	Unit
Lo	-2 – 2	-1	—
Hi	-2 – 2	1	—

Constant 0 inputs

DC source: emits a fixed Value every sample with no input. Patch it into a maths block as a constant operand, into a parameter mod input as a manual offset, or sum it onto a signal as bias. The simplest way to add a tunable number to the graph.

Param	Range	Default	Unit
Value	-1 – 1	1	—

Crossfade 3 inputs

Blends from in0 to in1 by a fade position of Mix plus the in2 CV (clamped 0..1). Mode 0 is a linear (constant-sum) fade; Mode 1 is an equal-power (constant-energy) fade for click-free audio crossfades. Patch in2 to automate the morph; leave it unpatched to set the blend by hand with Mix.

Param	Range	Default	Unit
Mix	0 – 1	0.5	—
Mode	Linear · Equal Power		—

Select 3 inputs

Two-way signal switch: routes in0 when the in2 selector (plus the Threshold) is below 0.5 and in1 when it is at or above, so a gate or LFO can hard-flip between two sources. Unlike Crossfade there is no blend - it is a clean A/B route for muxing oscillators or CV paths.

Param	Range	Default	Unit
Threshold	-1 – 1	0	—

Power 1 input

Sign-preserving exponent: outputs $\text{sign}(\text{in0}) * |\text{in0}|^{\text{Exp}}$, so it reshapes a normalised signal's curve without changing its polarity or its 0 and +/-1 endpoints. Exp > 1 bends a CV toward the extremes (snappier envelopes), Exp < 1 expands the middle (gentler control feel).

Param	Range	Default	Unit
Exp	0.1 – 8	2	—

Sign 1 input

Sign extractor with a dead zone: outputs -1, 0, or +1 depending on whether in0 sits below, inside, or above a +/-Dead band around zero. Turns any signal into a bipolar gate, derives direction from a slope, or squares an oscillator into a pulse. Dead at 0 gives a pure two-state comparator.

Param	Range	Default	Unit
Dead	0 – 1	0	—

Quantize 1 input

Rounds in0 to the nearest of Steps evenly-spaced levels across the [-1, 1] range, turning a smooth signal into a stepped one. Drop the resolution of a sweep into discrete plateaus, build sample-rate-style bit-crush on CV, or snap a glide into a staircase. Steps at 2 gives a bipolar square; higher Steps gives finer stairs.

Param	Range	Default	Unit
Steps	2 – 64	8	—

Fold 1 input

Triangle wavefolder: when $|\text{in0} * \text{Drive}|$ exceeds Threshold the excess is reflected back inward, repeatedly, so a loud signal folds onto itself and sprouts bright upper harmonics (Serge / Buchla west-coast timbre). Unlike a clipper it keeps adding partials with more Drive instead of flattening into a square.

Param	Range	Default	Unit
Drive	1 – 16	1	—
Threshold	0.1 – 1	1	—

Wrap 1 input

Phase-style wrap: folds in0 back into the [-Range, Range] window by wrapping (modulo) rather than clamping, so a value leaving one edge re-enters from the opposite edge. Ideal for keeping a phasor or free-running ramp bounded, or for hard sync-style discontinuities.

Param	Range	Default	Unit
Range	0.1 – 2	1	—

Modulo 1 input

Euclidean remainder: outputs in0 wrapped into $[0, \text{Mod})$, the non-negative remainder of $\text{in0} / \text{Mod}$. Generate ramps from a rising counter, build polyrhythmic resets, or derive a phasor from accumulated CV. Differs from Wrap, which keeps a bipolar +/-Range window centred on zero.

Param	Range	Default	Unit
Mod	0.01 – 2	1	—

Compare 1 input

Schmitt comparator: outputs +1 (high) when in0 rises above Threshold + Hyst and -1 (low) when it falls below Threshold - Hyst, holding its state in between so noise near the edge cannot chatter. Turns an audio or control signal into a clean bipolar gate; Hyst at 0 is an instantaneous comparator.

Param	Range	Default	Unit
Threshold	-1 – 1	0	—
Hyst	0 – 0.5	0.02	—

Logic 2 inputs

Boolean gate combiner: treats in0 and in1 as logic highs when they exceed 0.5 and applies AND, OR, XOR, NAND, NOR, or XNOR (Op), emitting +1 for true and -1 for false. Build trigger conditioners, gate masks, and rhythmic combinations from clocks, gates, and square LFOs.

Param	Range	Default	Unit
Op	AND · OR · XOR · NAND · NOR · XNOR	—	—

Map 1 input

Affine range mapper: rescales in0 from its [In Lo, In Hi] range onto a [Out Lo, Out Hi] range, then optionally clamps to the output bounds. The general-purpose glue for matching signal ranges - convert a 0..1 envelope to a +/-1 CV, narrow an LFO's swing, or invert a range by setting Out Lo > Out Hi.

Param	Range	Default	Unit
In Lo	-2 – 2	-1	—
In Hi	-2 – 2	1	—
Out Lo	-2 – 2	0	—
Out Hi	-2 – 2	1	—
Clamp	Off · On	—	—

Sum 4 inputs

Summing node: adds inputs in 1..4 into a single output with no per-input gain - the quick way to merge oscillators, parallel chains or control signals without wiring up a full Mixer.

PhaseInvert 1 input

Flips the polarity of the signal, negating every sample when engaged and passing it untouched otherwise. Invert toggles the inversion on or off. Use it to correct out-of-phase tracks or to null one source against another.

Param	Range	Default	Unit
Invert	0 – 1	1	—

DCBlock 1 input

Removes DC offset and sub-sonic bias with a one-pole high-pass that subtracts the previous input and feeds back a 0.9995 coefficient, passing audio while draining any constant component. It takes no parameters and always runs at the same fixed corner. Place it after asymmetric distortion or rectification stages to keep the signal centred.

Dither 1 input

Bit-depth reducer with triangular (TPDF) dither: builds a quantization step from Bits, adds triangular noise from two xorshift draws scaled by Amount, then rounds to the step. Bits sets the target word length from 8 to 24 and Amount scales the dither level. The dither decorrelates quantization error so low-level detail survives reduction instead of distorting.

Param	Range	Default	Unit
Bits	8 – 24	16	—
Amount	0 – 1	1	—

CVBridge 1 input

Control-voltage utility that conditions a single input value in one of three Mode settings: attenuvert (scale by Scale plus Offset), note-to-1V/oct pitch conversion, or bipolar-to-unipolar remap. Scale provides positive or negative gain including inversion, and Offset shifts the result up or down. Useful for routing and rescaling modulation or pitch control between modules.

Param	Range	Default	Unit
Mode	0 – 2	0	—
Scale	-2 – 2	1	—
Offset	-1 – 1	0	—

Meter 1 input

Analysis meter that tracks the running mean-square of the input and outputs its RMS level as a scalar control value. Smooth sets the one-pole averaging coefficient, with higher values giving slower, steadier readings and lower values reacting faster to transients. Used to derive a level signal for metering or to drive other modules from program loudness.

Param	Range	Default	Unit
Smooth	0 – 0.999	0.99	—

NullTest 2 inputs

Null tester that subtracts the second input from the first and outputs the residual difference, scaled by Gain. When two signals are identical the output is silence, and any audible residual reveals the difference between them. Gain amplifies that residual so small mismatches become easy to hear, making this a tool for verifying bit-for-bit equivalence between processing chains.

Param	Range	Default	Unit
Gain	0 – 4	1	—

Spectral

12 modules

SpecGate 1 input

Spectral gate: runs a streaming STFT and, per frame, zeroes any FFT bin whose magnitude falls below Threshold relative to the loudest bin. Threshold sets that cutoff fraction (0 passes everything, 1 keeps only the peak); Mix blends the gated spectrum against the dry input. Keeps only the strongest partials, thinning noise and weak harmonics into a sparser, cleaner tone.

Param	Range	Default	Unit
Threshold	0 – 1	0.2	—
Mix	0 – 1	1	—

SpecFreeze 1 input

Spectral freeze: while Freeze is off it tracks each FFT bin's magnitude per frame; when Freeze is on it holds those captured magnitudes (keeping the live phase) to sustain the last spectrum indefinitely. Freeze is the on/off latch and Mix blends the held spectrum against the dry input. Turns a transient moment into a steady drone or pad.

Param	Range	Default	Unit
Freeze	0 – 1	0	—
Mix	0 – 1	1	—

SpecFilter 1 input

Per-bin spectral filter: runs a streaming STFT and zeroes whole FFT bins on one side of a cutoff bin for a brick-wall response in the frequency domain. Cutoff is the bin index of the edge and Mode selects low-pass (zero bins above) or high-pass (zero bins below); Mix blends against the dry input. The hard bin cut gives a far steeper, ringing-free roll-off than an analog filter.

Param	Range	Default	Unit
Cutoff	1 – 512	256	bin
Mode	0 – 1	0	—
Mix	0 – 1	1	—

SpecComp 1 input

Spectral compressor: per STFT frame it finds the loudest bin, then attenuates every bin above Threshold (relative to that peak) back toward the threshold. Threshold sets the relative knee and Amount scales how hard over-threshold bins are pulled down; Mix blends against the dry input. Evens out spectral dynamics, taming dominant partials for a flatter, denser spectrum.

Param	Range	Default	Unit
Threshold	0 – 1	0.3	—
Amount	0 – 1	0.6	—
Mix	0 – 1	1	—

SpecDelay 1 input

Spectral echo: per STFT frame each bin's magnitude is summed with its previous-frame value scaled by Feedback, and that running magnitude is fed back (live phase retained). Feedback (0..0.95) sets how much each bin's energy persists and decays across frames; Mix blends against the dry input. Produces a smeared, frequency-domain tail closer to a freeze/reverb than a discrete time echo.

Param	Range	Default	Unit
Feedback	0 – 0.95	0.5	—
Mix	0 – 1	1	—

SpecBlur 1 input

Spectral blur: per STFT frame each bin's magnitude is a Blur-weighted mix of its previous value and the current one, smearing magnitudes over time while keeping live phase. Blur (0..1) sets the smoothing amount -- higher values hold past energy longer; Mix blends against the dry input. Softens transients and stretches the sound into a sustained pad.

Param	Range	Default	Unit
Blur	0 – 1	0.5	—
Mix	0 – 1	1	—

SpecMorph 1 input

Spectral smoothing: per STFT frame it runs a one-pole average up the bins, replacing each magnitude with a Smooth-weighted blend of the running value and the current bin (live phase retained). Smooth (0..1) sets how strongly the envelope is smeared across frequency -- higher values flatten spectral detail; Mix blends against the dry input. Blurs sharp formants into a softer, more uniform tone.

Param	Range	Default	Unit
Smooth	0 – 1	0.5	—
Mix	0 – 1	1	—

SpectralReso 1 input

Spectral resonator: per STFT frame each bin is compared with its two neighbors; local peaks are boosted by up to $1 + \text{Amount} \times 2$ while non-peak bins are cut by up to $\text{Amount} \times 0.5$. Amount (0..1) sets both the peak boost and the valley attenuation; Mix blends against the dry input. Sharpens the spectrum into a more resonant, harmonically pronounced, vowel-like tone.

Param	Range	Default	Unit
Amount	0 – 1	0.5	—
Mix	0 – 1	1	—

FormantShift 1 input

Formant shift: per STFT frame it captures the magnitude envelope, resamples it by a ratio of $2^{(\text{Shift}/12)}$, then re-weights each bin to impose the shifted envelope while leaving the underlying partials in place. Shift (+/-12 semitones) moves the formants up or down, so pitch stays fixed while timbre changes; Mix blends against the dry input. Produces vocal gender/size effects without affecting the played note.

Param	Range	Default	Unit
Shift	-12 – 12	0	st
Mix	0 – 1	1	—

Soothe 1 input

Dynamic resonance suppressor: per STFT frame it tracks a smoothed local average up the bins and pulls any bin above that average down toward it, scaled by Amount. Amount (0..1) sets how aggressively over-average peaks are tamed (0 leaves them, 1 flattens them to the average); Mix blends against the dry input. Reduces harsh resonant spikes for a smoother, more even spectrum.

Param	Range	Default	Unit
Amount	0 – 1	0.6	—
Mix	0 – 1	1	—

MatchEQ 1 input

Matching EQ: with Capture on it learns each bin's magnitude as a reference curve; with Capture off it scales every bin toward that stored reference per frame. Capture latches learn vs apply, Amount (0..1) blends between no change and a full match (bin gain clamped to 8x); Mix blends against the dry input. Reshapes the input's tonal balance to resemble a previously captured reference.

Param	Range	Default	Unit
Capture	0 – 1	0	—
Amount	0 – 1	0.7	—
Mix	0 – 1	1	—

LinearEQ 1 input

Linear-phase 3-band EQ: per STFT frame it applies a single magnitude gain to each FFT bin grouped into low (below half/8), mid, and high (from half/2) bands, so the output has zero phase shift. Low, Mid, and High set each band's gain in dB (converted via $10^{(dB/20)}$); Mix blends against the dry input. The linear-phase response avoids the phase smearing of analog EQ, useful on transient or stereo material.

Param	Range	Default	Unit
Low	-18 – 18	0	dB
Mid	-18 – 18	0	dB
High	-18 – 18	0	dB
Mix	0 – 1	1	—

Stereo

11 modules

StereoWidth 2 inputs

Mid/side stereo widener: decomposes the L/R pair into mid (sum) and side (difference), scales the side by Width, then recombines. Width below 1 narrows toward mono and above 1 (up to 2) exaggerates the stereo spread. Out picks L or R, so one instance emits each recombined channel; useful for tightening or opening a stereo bus.

Param	Range	Default	Unit
Width	0 – 2	1	—
Out	0 – 1	0	—

MidSide 2 inputs

Mid/side processor with independent gain on each component: splits L/R into mid and side, applies MidGain and SideGain in dB, then recombines. Lowering MidGain or raising SideGain widens the image while the reverse focuses the centre. Out picks L or R for the recombined channel; handy for level-matching centre versus stereo content.

Param	Range	Default	Unit
MidGain	-24 – 12	0	dB
SideGain	-24 – 12	0	dB
Out	0 – 1	0	—

MonoMaker 2 inputs

Bass mono-maker: a one-pole low-pass at Freq splits the spectrum so content below the crossover is summed to mono while the highs stay in their own channel. Freq sets the crossover from 20 to 500 Hz. Out picks L or R, supplying that channel's highs over the shared mono low end; tightens low-frequency phase for vinyl and club playback.

Param	Range	Default	Unit
Freq	20 – 500	150	Hz
Out	0 – 1	0	—

Haas 1 input

Haas widener for a mono source: feeds the input into a delay line and outputs either the dry signal or a copy delayed by a few milliseconds. Delay sets the offset from 0 to 40 ms, exploiting the precedence effect to throw the image to one side. Out picks L (dry) or R (delayed) so the pair forms a wide stereo spread from one mono input.

Param	Range	Default	Unit
Delay	0 – 40	15	ms
Out	0 – 1	0	—

Panner 1 input

Constant-power panner for a mono source: maps Pan (-1 to +1) onto a sine/cosine gain pair so total power stays even across the field. Pan at 0 centres the source; the +/-1 extremes hard-pan it. Out picks L or R, each instance applying the matching sine or cosine leg of the pan law.

Param	Range	Default	Unit
Pan	-1 – 1	0	—
Out	0 – 1	0	—

AutoPan 1 input

LFO auto-panner for a mono source: a sine LFO driven from the shared transport time sweeps a constant-power pan automatically. Rate sets the sweep speed from 0.05 to 10 Hz and Depth scales how far the source travels. Out picks L or R, and because the LFO is phase-coherent across instances the two outputs track in opposition for stable motion.

Param	Range	Default	Unit
Rate	0.05 – 10	1	Hz
Depth	0 – 1	1	—
Out	0 – 1	0	—

Stereoize 1 input

Mono-to-stereo decorrelator: runs the input through a four-stage all-pass chain to phase-shift it without changing tone, then blends that toward the dry signal by Amount. Amount controls how much decorrelated signal is mixed in. Out picks L (dry) or R (decorrelated) so the stereo pair widens a mono source while keeping a centred reference.

Param	Range	Default	Unit
Amount	0 – 1	0.7	—
Out	0 – 1	0	—

StereoImager 2 inputs

Frequency-dependent imager: splits the side signal at Freq with a one-pole low-pass and scales only the high side band by Width, leaving the low side untouched. Width sets the high-band spread and Freq sets the split point from 100 to 2000 Hz. Out picks L or R for the recombined channel; widens air and detail while keeping the low end mono-safe.

Param	Range	Default	Unit
Width	0 – 2	1.3	—
Freq	100 – 2000	400	Hz
Out	0 – 1	0	—

FreqPanner 2 inputs

Frequency-split panner: sums the input to mono, low-passes it at Freq, and applies a constant-power pan to only the high band while the low band stays centred. Freq sets the split from 100 to 4000 Hz and Pan positions the highs from -1 to +1. Out picks L or R, applying the matching pan-law leg to the highs over the shared centred lows.

Param	Range	Default	Unit
Freq	100 – 4000	800	Hz
Pan	-1 – 1	0.5	—
Out	0 – 1	0	—

Crossfeed 2 inputs

Headphone crossfeed: low-passes the opposite channel and bleeds it into this one by Amount, emulating how each ear hears both speakers in a room. Amount sets the bleed level from 0 to 1. Out picks L or R, choosing which channel is the destination and which is the filtered source; eases the hard-panned separation that fatigues on headphones.

Param	Range	Default	Unit
Amount	0 – 1	0.3	—
Out	0 – 1	0	—

Doubler 1 input

Vocal doubler: writes the input to a delay line and reads back a single detuned, delayed copy whose time drifts via a slow c.time-phased LFO for a natural double-tracked feel. Detune scales the drift depth, Delay sets the base delay time, and Out shifts the copy slightly shorter or longer (0.9x or 1.1x of the base) to place it as the left or right voice. Layer two instances on opposite Out settings for a widened stereo double.

Param	Range	Default	Unit
Detune	0 – 30	12	cents
Delay	10 – 50	25	ms
Out	0 – 1	0	—

Wavetables

Built-in wavetable names selectable in the Wt / Wavetable oscillator.

Basic Shapes Sine to Saw Harmonic Sweep PWM Sync Sweep Formant